



RFL-MMTLT

**Motorized Micro-strip Transmission
Line Trainer**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



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- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
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- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

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This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

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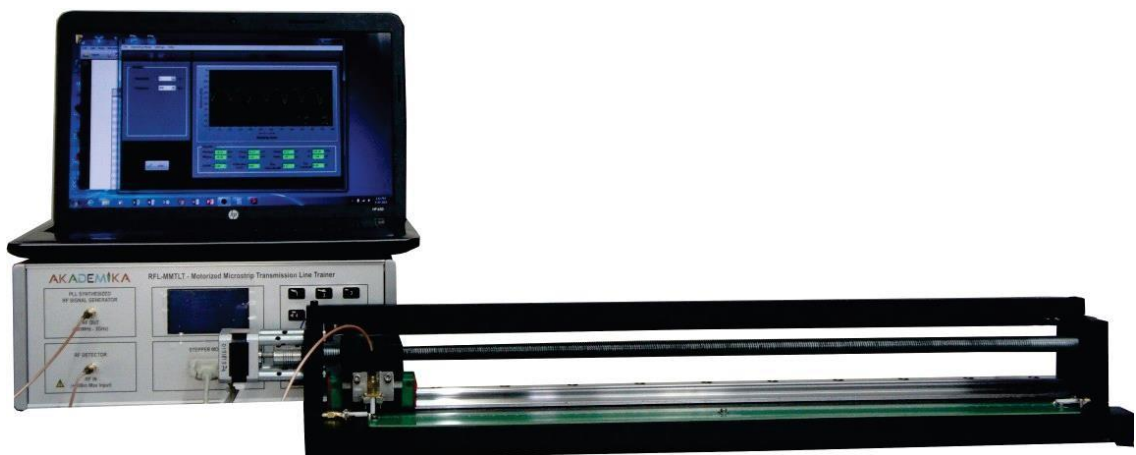
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RFL-MMTLT: MOTORIZED MICROSTRIP TRANSMISSION LINE TRAINING

INTRODUCTION

RFL-MMTLT is designed to understand the transmission line theory. The setup primarily focuses on minima and maxima measurement in standing wave for various load condition as well as various frequencies. The sensor probe moves along the transmission line using motorized mechanism. User friendly GUI of PC based Software allows parameter settings, control as well as analysis of acquired data in graphical and tabular manner. It allows various measurements such as Return loss, VSWR, Impedance, Power etc. These parameters help students to learn the characteristics of transmission line terminated with various impedance condition. MMTLT operates over frequency range from 250MHz to 1500MHz in UHF band. A standalone mode of operation allows all the operations which are possible with PC based software.



Features:

- 50 Ω Microstrip Transmission line
- 1mm Sensing Probe Resolution
- Standing wave pattern at various frequencies
- Guided Wavelength Measurement
- Characterization for given LOADS.
- Characterization for Unknown LOADS Vs Frequency
- S-Parameters (S11, S21, S12, S22) Measurement
- Analysis of reflection coefficients, VSWR and Impedance for LOAD, OPEN and SHORT termination
- Smith Chart Analysis

TECHNICAL SPECIFICATIONS

1. Microstrip Transmission Line

1.	Frequency	250MHz TO 1500MHz
2.	Type of device	Passive
3.	3rd and 7th Harmonic level	negligible
4.	Characteristics impedance	50 ohms
5.	Source Type	PLL Synthesized
6.	Power	+9dBm
7.	Resolution	150 KHz
8.	Copper thickness	1oz (35um)
9.	Power handling capacity	+30dBm
10.	Detector Type	Dual Log Type
11.	Connector	SMA female, (Jack)
12.	Length of Transmission line	590mm
13.	Sensing Probe Resolution	1mm
14.	Probe Output	dBm/ uW/ mV

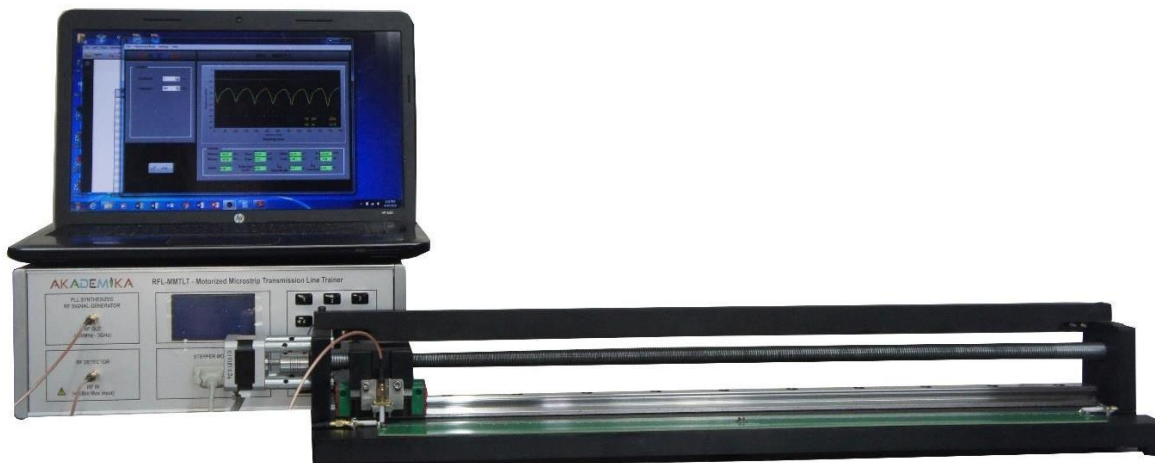
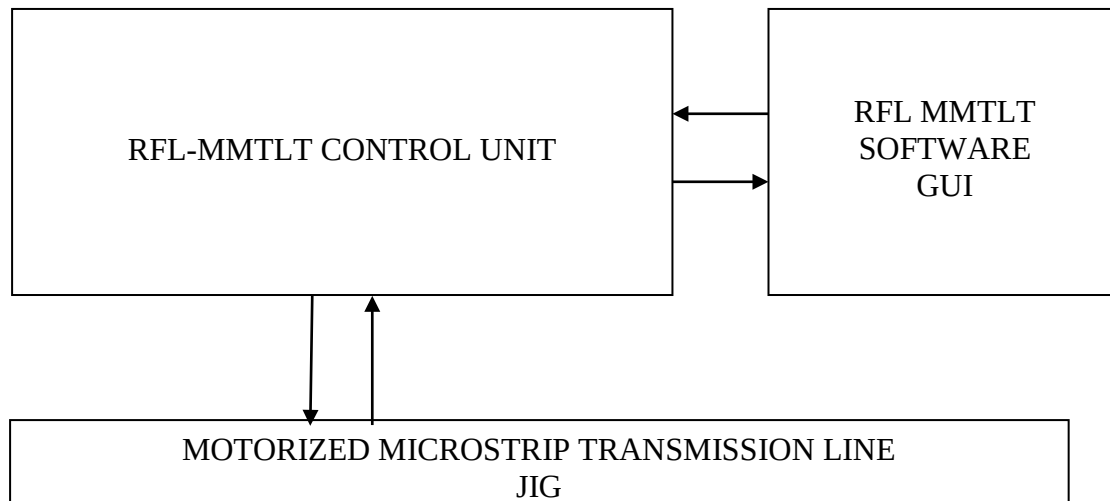
2. Directional Coupler

1.	Frequency	20 to 3000 MHz
2.	Number of port	3
3.	Result/graph plotting	S11, Z, VSWR
4.	Min VSWR detection	1.18:1
5.	Impedance	50 Ω
6.	Minimum return loss(S11)	-30dB
7.	Power handling	1W
8.	Temperature range	0 to 55
9.	Measurement error in terms of frequency	± 2.56 dB
10.	Measurement error in terms of power	± 20 MHz
11.	Directivity	20dB

Deliverables

Sr. No.	Description	Quantity
1	Transmission line - JIG	01
2	Transmission line –Control Unit	01
3	SMA (M) to SMA (M) 50-ohm Coaxial Cable-2meter	01
4	SMA (M) to SMA (M) 50-ohm Coaxial Cable- 1 meter	02
5	Directional Coupler	01
6	USB A Male to USB B Male cable	01
7	Power Cord	01
8	9 pin D type male to female cable	01
9	9 pin D Type Male to DIN (Male) Connector Cable	01
10	Standard Terminations (Open, Short , Load)	01
11	Software on Pen Drive	01
12	Experimental manual	01

FUNCTIONAL BLOCK



1	Transmission line JIG with motor	
2	RFL-MMTLT Control Unit	
3	PC based control using software	
4	Directional Coupler	

RF BASICS

The technology involving the generation, the manipulation, the transmission and the reception of radio waves, and the use of these to transmit information (both in analog and digital way), is called RF technology or, nowadays, *wireless* technology. The literal meaning of RF is *Radio Frequency*, but this term is often use in the figurative sense of “anything related to electromagnetic signals”.

An electromagnetic or RF signal is a controlled variation of electrical energy over time. The simplest example of such a variation is when the intensity of the electric field grows to some maximum, then goes back to zero, becoming negative and reaching some minimum value, before returning back to zero and then starting the process again, following the shape of the trigonometric function called *sine*. The *sine wave* is the basic example of a signal that can be generated, transmitted and received with RF equipment. It may be characterized by its *frequency* and its *amplitude*:

- the frequency is the number of times a signal goes through a complete “up and down” cycle in one second of time, and is measured in *Hertz* (being 1 Hz the frequency of a signal that makes a complete cycle every second);
- the amplitude of a signal is the difference between its maximum and its minimum value during one cycle. It is measured with the same unit used for the intensity of the electric field, the *Volt* (V), and is related with the strength, or power, of the signal.

The units used to measure frequency and amplitude may become inadequate when the numeric values of such quantities become very small or very big, so that a different notation is needed, called the scientific notation, that use the powers of ten to multiply the values. It gives also standard names to the many prefixes associated with these multiplying factors.

Some are listed here:

milli (m)	1/1000	10 to the power of -3 ex.: 1 mV
micro (μ)	1/1000000	10 to the power of -6 ex.: 1 μV
kilo (k)	1000	10 to the power of 3 ex.: 1 kHz
Mega (M)	1 million	10 to the power of 6 ex.: 1 MHz
Giga (G)	1 billion	10 to the power of 9 ex.: 1 GHz

Frequency, bands and bandwidth

The concept of frequency is the key to understanding RF, because almost everything in the RF world is frequency-dependent. Signals can be distinguished on the basis of their different frequencies, and RF devices are almost always characterized by the range of frequencies they can generate, measure or handle.

The range of frequencies used is also the most common way to distinguish one wireless application from another. This is achieved using the concept of *frequency band* that is a standard way to name a specific range of frequencies. A list of RF bands of common use is given here:

HF	High Frequency	3 MHz - 30 MHz
VHF	Very High Frequency	30 MHz - 300 MHz
UHF	Ultra High Frequency	300 MHz - 3 GHz

SHF	Super High Frequency	3 GHz - 30 GHz
EHF	Extra High Frequency	30 GHz - 300 GHz

While for the microwave range a more detailed band list is this one:

L band	1GHz - 2 GHz
S band	2GHz - 4GHz C
band	4GHz - 8GHz X
band	8GHz - 12GHz
Ku band	12GHz - 18GHz K
band	18GHz - 27GHz Ka
band	27GHz - 40GHz

The bands we are mostly interested in for this handbook are the ISM (Industrial, Scientific and Medical) radio bands that were originally reserved internationally for non-commercial use of RF electromagnetic fields for industrial, scientific and medical purposes. The ISM bands are defined by the ITU-T in S5.138 and S5.150 of the Radio Regulations and are used for license-free error-tolerant digital communications applications such as wireless LANs and Bluetooth:

"900MHz" ISM band	902MHz - 928MHz (Americas only)
"2.4GHz" ISM band	2400MHz - 2483.5MHz
"5GHz" ISM band	5725MHz - 5875MHz

Many other RF bands and possible classifications and definitions exist, that we will not consider here.

Together with the concept of frequency, there is another useful one: the *bandwidth*. While in the telecommunications world the term bandwidth has a general meaning of how much information can be carried in a given period of time (usually a second) over a wired or wireless communications link, in the RF world the bandwidth is the width of the range of frequencies that an electronic signal occupies on a given transmission medium. Any RF signal has a bandwidth, expressed in terms of the difference between the highest-frequency signal component and the lowest-frequency signal component. A typical voice transmission like a phone call has a bandwidth of approximately 3 kHz; a stereo musical signal like a FM radio broadcast has a bandwidth of 200 KHz; an analog television (TV) broadcast video signal has a bandwidth of 6MHz.

Wavelength

The wavelength is the distance a radio wave will travel during one cycle. The formula which relates the wavelength to the frequency is the following:

$$\lambda = \frac{c}{f}$$

Where λ is the wavelength, expressed in meters, c is the speed of light, which is exactly 299793,07 meters/second, and f is the frequency.

Following is a table with the wavelength for different ISM frequencies:

Frequency	Wavelength
900 MHz	0.33 m
2.4 GHz	0.125 m
5.0 GHz	0.06 m

The speed of propagation at a certain frequency in a coaxial cable is slower than in air, so the wavelength is shorter. The velocity of propagation of electromagnetic waves in coaxial cables is usually given as a percentage of the free space velocity, and it is different for different types of coaxial cables, and is called Velocity Factor (VF).

Amplitude and power

The amplitude of an RF signal, measured in Volts, is not of common use in the RF world, while the related concept of *power* of a signal is much more used and useful.

The power of a constant low-frequency signal is measured in *Watts*, symbolized by *W*, and is the product of the amplitude (or voltage, measured in Volts) and the current (measured in Amperes). One watt is also the power resulting from an energy dissipation, conversion, or storage process equivalent to one Joule per second. At high frequencies, in which energy is stored and released (as well as dissipated or converted), the relation between power, amplitude and current is more complex.

In a DC circuit, a source of *E* volts, delivering *I* Amperes, produces *P* Watts according to the formula:

$$P = E I$$

When a current of *I* Amperes (A) passes through a *resistance* of *R* Ohms (Ω), then the power in Watts dissipated or converted by that component is given by:

$$P = I^2 R$$

When a potential difference of *E* Volts appears across a component having a resistance of *R* Ohms, then the power in Watts dissipated or converted by that component is given by:

$$P = E^2 / R$$

In a DC circuit, power is a scalar (one-dimensional) quantity. In the general RF case, the determination of power requires two dimensions, because RF power is a vector quantity. Assuming there is no reactance (opposition to RF but not to DC) in an RF circuit, the power can be calculated according to the above formulas for DC, using root-mean-square values for the alternating current and voltage. If reactance exists, some power is alternately stored and released by the system. This is called apparent power or reactive power. The resistance dissipates power as heat or converts it to some other tangible form; this is called true power. The vector combination of reactance and resistance is known as impedance.

Decibels and dB math

The *decibel* (abbreviated as *dB*) is a logarithmic expression of the ratio between the power, voltage, or current of two signals. A decibel is one-tenth of a Bel, a seldom-used unit named for Alexander Graham Bell.

Suppose a signal has a power of *P1* watts, and a second signal has a power of *P2* watts. Then the power ratio in decibels, symbolized *PdB*, is:

$$P_{dB} = 10 \log_{10} (P_2 / P_1)$$

Decibels can also be calculated in terms of the effective voltage if the load impedance remains constant. Suppose a signal has an RMS (root mean- square) voltage of V_1 across a load, and a second signal has an RMS voltage of V_2 across another load having the same impedance. Then the voltage amplitude difference in decibels, symbolized V_{dB} , is:

$$V_{dB} = 20 \log_{10} (V_2 / V_1)$$

When a decibel figure is positive, then the second signal is stronger than the first signal and the power ratio is called *gain*. When a decibel figure is negative, then the second signal is weaker than the first signal and the power ratio is called *loss*. In amplifiers, the gain, also called the amplification factor, is often expressed in decibels.

For example, if a signal at the output of an amplifier is 100 times bigger than the signal at its input, the amplifier has a gain of 100 or, using the definition of decibel, of 20 dB. Some useful values to remember are:

+3 dB	means two times bigger
+10 dB	means ten times bigger
- 3 dB	means one half
-10 dB	means one tenth

To express power using decibels we need a specific power to be assumed as a reference. In the RF world the common standard is to refer powers to 1 mW (0.001 Watts). Such power ratio, expressed in decibels, is called dBm. Following the definition of dB, we have:

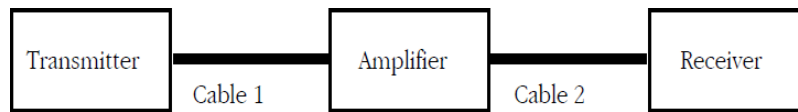
$$P_{dBm} = 10 \log_{10} (P_{Watts} / 1 \text{ mW})$$

A table of correspondence between Watts and dBm is given here:

1 μ W	= -30 dBm
10 μ W	= -20 dBm
100 μ W	= -10 dBm
1 mW	= 0 dBm
10 mW	= 10 dBm
100 mW	= 20 dBm
1 W	= 30 dBm
10 W	= 40 dBm
100 W	= 50 dBm

The advantage of using decibels instead of Watts to express the power of a signal along an *RF chain* (a chain composed by RF devices like transmitters, receivers, cables, amplifiers, attenuators, measurement instruments, loads, etc.) is that instead of dividing or multiplying powers to take care of amplifications and attenuations, we just add or subtract the gains and the losses expressed in decibels. An example follows:

Transmitter Amplifier Receiver



Using amplification and attenuation factors and expressing the powers in Watts, we obtain the value of the power at the receiver's input in this way:

$$P_{\text{Receiver}} = P_{\text{Transmitter}} \times (1 / \text{Attenuation}_{\text{Cable1}}) \times \text{Amplification} \times (1 / \text{Attenuation}_{\text{Cable2}})$$

If we use dB to express the gains and the losses and dBm to express the powers, the calculation becomes a simple addition:

$$P_{\text{Receiver}} = P_{\text{Transmitter}} + \text{LOSS}_{\text{Cable1}} + \text{Gain}_{\text{Amplifier}} + \text{LOSS}_{\text{Cable2}}$$

The procedure of adding gains and losses to obtain the resulting power for an RF chain is very common and is called *Power Budget Calculation*.

Transmission Lines

Introduction

The transmitter that generates the RF power to drive the antenna is usually located at some distance from the antenna terminals. The connecting link between the two is the *RF transmission line*. Its purpose is to carry RF power from one place to another, and to do this as efficiently as possible. From the receiver side, the antenna is responsible for picking up any radio signals in the air, and passing them to the receiver with the minimum amount of distortion so that the radio has its best chance to decode the signal. For these reasons, the RF cable has a very important role in radio systems: it must maintain the integrity of the signals in both direction.

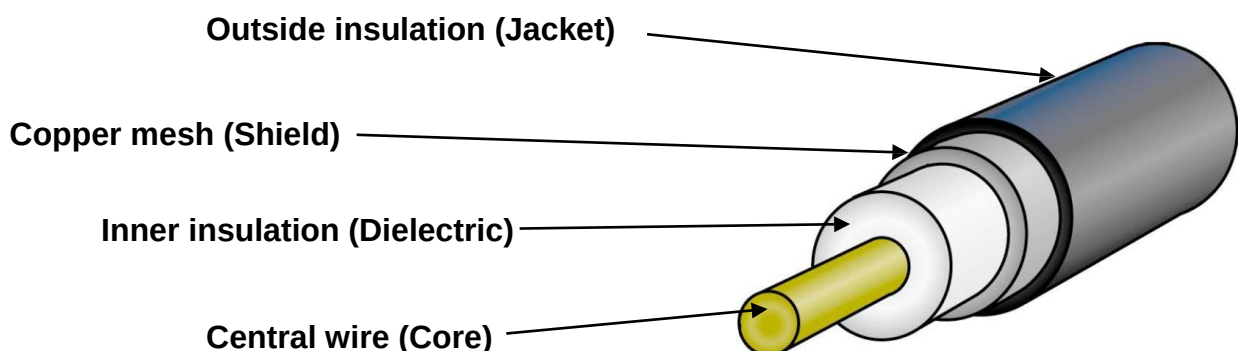
The need for efficient transfer of RF power is one of the main reasons behind the use of transmission lines. At low frequencies where the wavelength of the signals are much larger than the length of the circuit conductors, a simple wire is very useful for carrying power. Current travels down the wire easily, and voltage and current are the same no matter where we measure along the wire.

At high frequencies however, the wavelength of signals of interest are comparable to or much smaller than the length of conductors. In this case, power transmission can best be thought of in terms of traveling waves.

There are two main categories of transmission lines: *cables* and *waveguides*.

Cables

RF cables are, for frequencies higher than HF, almost exclusively *coaxial cables* (or "coax" for short, derived from the words "of common axis"). Coax cables have a core wire, surrounded by a non-conductive material (which is called dielectric or insulation), and then surrounded by an encompassing shielding which is often made of braided wires. The dielectric keeps the core and the shielding apart. Finally, the coax is protected by an outer shielding which will generally be a PVC material. The inner conductor carries the RF signal and the outer shield is there to keep the RF signal from radiating to the atmosphere and to stop outside signals from interfering with the signal carried by the core. Another interesting fact is that the electrical signal always travels along the outer layer of the central conductor: the larger the central conductor, the better signal will flow. This is called the "skin effect".



In the following table the diameters of Core, Dielectric, Shield and Jacket of the most popular Coax cables can be found.

Cable Type	Core (mm)	Dielectric (mm)	Shield (mm)	Jacket (mm)
RG-58	0.9	2.95	3.8	4.95
RG-213	2.26	7.24	8.64	10.29
LMR-400	2.74	7.24	8.13	10.29
3/8" LDF	3.1	8.12	9.7	11

Even though the coaxial construction is good at containing the signal on the core wire, there is some resistance to the electrical flow: as the signal travels down the core, it will fade away. This fading is known as attenuation, and is measured in dB/m. The rate of attenuation is a function of the signal frequency and the physical construction of the cable itself. And a table of these values can be found in the next chapter. Obviously, we need to minimize the cable attenuation as much as possible, keeping the cable very short and using high quality cables.

Practical Hints: How to choose the proper cable

- “The shorter the better!”: the first rule when you have to place a cable is to try to keep it as short as possible. The power loss is not linear, so doubling the cable length means that you are going to lose much more than twice the power. In the same way, halvening the cable length gives you more than twice the power at the antenna. The best solution is to place the transmitter as close as possible to the antenna, even when this means placing it on the top of a mast.

- “The cheaper the worse!”: the second golden rule is that the money you invest in buying a good quality cable is a bargain. Cheap cables are intended to be used at low frequencies, not higher than VHF. Microwaves require the highest quality cables; all other options are nothing else than a dummy load.

- always avoid *RG-58*. It is intended for thin Ethernet networking, CB or VHF radio, not for microwave.

- always avoid *RG-213*. It is intended for CB and HF radio. The cable diameter does not imply a high quality, or low attenuation.

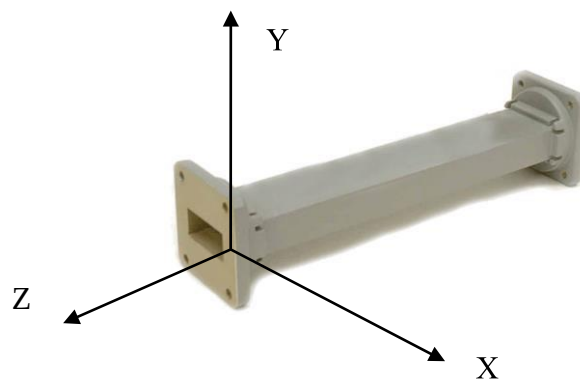
- always use “*Helix*” (also called “*Foam*”) cables for connecting the transmitter to the antenna, and Semi-rigid cables to interconnect the other devices in the RF chain (i.e. instrumentation). Semi-rigid cables consist of a solid inner conductor (usually copper or silver/copper-plated steel), a solid PTFE dielectric and a solid copper outer pipe. They are sealed, and their rigid construction means that they cannot flex. However, the loss is still relatively high due to the small diameter, and they are therefore rarely used as a feeder to an antenna. Helix cables are basically larger diameter flexible versions of Semi-rigid cables, with a corrugated solid outer conductor to enable them to flex more easily. They can be built in two ways: using air or foam as dielectric. The first solution is the most expensive, guarantees the minimum loss but is very difficult to handle. The second solution is more loss but less expensive and easier to install. A special procedure is required when soldering connectors, in order to maintain the foam dielectric dry and uncorrupted.

- never step over a cable, never bend it too much, never try to unplug a connector pulling directly the cable. All those behaviors may change the mechanical characteristic of the cable and therefore its impedance, shortcut the inner conductor with the shield, or even break the line. Those problems are difficult to track and recognize and can lead to unpredictable behavior of the radio link.

Waveguides

Above 2 GHz, the wavelength is short enough to allow practical, efficient energy transfer by different means. A waveguide is a conducting tube through which energy is transmitted in the form of electromagnetic waves. The tube acts as a boundary that confines the waves in the enclosed space. The skin effect prevents any electromagnetic effects from being evident outside the guide. The electromagnetic fields are propagated through the waveguide by means of reflections against its inner walls, which are considered perfect conductors. The intensity of the fields is greatest at the center along the X dimension, and must diminish to zero at the end walls because the existence of any field parallel to the walls at the surface would cause an infinite current to flow in a perfect conductor. Waveguides, of course, cannot carry RF in this fashion.

The X, Y and Z dimensions of a rectangular waveguide can be seen in the following figure:



There are an infinite number of ways in which the electric and magnetic fields can arrange themselves in a waveguide for frequencies above the low cutoff frequency. Each of these field configurations is called a *mode*. The modes may be separated into two general groups. One group, designated *TM* (*Transverse Magnetic*), has the magnetic field entirely transverse to the direction of propagation, but has a component of the electric field in the direction of propagation. The other type, designated *TE* (*Transverse Electric*) has the electric field entirely transverse, but has a component of magnetic field in the direction of propagation. TM waves are sometimes called E waves, and TE waves are sometimes called H waves, but the TM and TE designations are preferred. The mode of propagation is identified by the group letters followed by two subscript numerals. For example, TE₁₀, TM₁₁, etc. The number of possible modes increases with the frequency for a given size of guide, and there is only one possible mode, called the *dominant mode*, for the lowest frequency that can be transmitted. In a rectangular guide the critical dimension is X. This dimension must be more than 0.5λ at the lowest frequency to be transmitted. In practice, the Y dimension usually is made about equal to $0.5 X$ to avoid the possibility of operation in other than the dominant mode. Cross-sectional shapes other than the rectangle can be used, the most important being the circular pipe. Much the same considerations apply as in the rectangular case. Wavelength dimensions for rectangular and circular guides are given

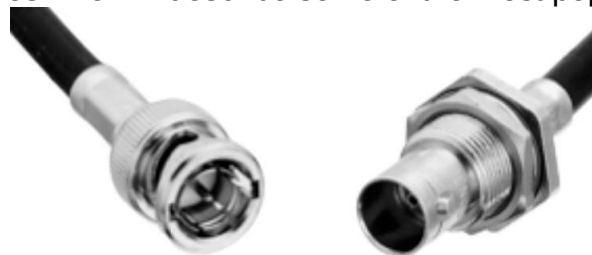
in the following table, where X is the width of a rectangular guide and r is the radius of a circular guide. All figures apply to the dominant mode.

Type of Guide	Rectangular	Circular
Cutoff Wavelength	$2X$	$3.41r$
Longest Wavelength transmitted with little attenuation	$1.6X$	$3.2r$
Shortest Wavelength before next mode becomes possible	$1.1X$	$2.8r$

Energy may be introduced into or extracted from a waveguide by means of either an electric or magnetic field. The energy transfer frequently is through a coaxial line. Two possible methods for coupling to a coaxial line are using the inner conductor of the coaxial line or through a loop. A probe which is simply a short extension of the inner conductor of the coaxial line can be oriented so that it is parallel to the electric lines of force. A loop can be arranged so that it encloses some of the magnetic lines of force. The point at which maximum coupling is obtained depends upon the mode of propagation in the guide or cavity. Coupling is maximum when the coupling device is in the most intense field. If a waveguide is left open at one end it will radiate energy. This radiation can be enhanced by flaring the waveguide to form a pyramidal horn antenna.

Connectors and Adapters

Connectors allow a cable to be connected to another cable or to a component of the RF chain. There is a wide variety of fittings and connectors designed to go with various sizes and types of coaxial lines. We will describe some of the most popular ones.



BNC connectors were developed in the late 40s, and BNC stands for Bayonet Neill Concelman after Amphenol's engineer Carl Concelman. The BNC product line is a miniature quick connect/disconnect connector. It features two bayonet lugs on the female connector, and mating is achieved with only a quarter turn of the coupling nut. BNC's are ideally suited for cable termination for miniature to subminiature coaxial cable (RG - 58 to RG - 179, RG - 316, etc.). They have acceptable performance up to few GHz.



Type N (Navy) connectors were originally developed during the Second World War. They are usable up to 18 GHz, and very common for microwave and available for almost all types of cable. Both the plug/cable and plug/socket joints are waterproof, providing an effective cable clamp.



SMA is an acronym for Subminiature version A and was developed in the 60s. 50Ω SMA connectors are precision, subminiature units that provide excellent electrical performance up to 18 GHz. These high-performance connectors are compact in size and mechanically have outstanding durability.



The *SMB* name derives from Subminiature B, and it is the second subminiature design. The SMB is a smaller version of the SMA with Snap-On coupling. It provides broadband capability through 4 GHz with a Snap-On connector design.



MCX connectors were introduced in the 80s. While the MCX uses identical inner contact and insulator dimensions as the SMB, the outer diameter of the plug is 30% smaller than the SMB. This series provides designers with options where weight and physical space are limited. MCX provides broadband capability though 6 GHz with a Snap-On connector design.



The *MMCX* series is also called MicroMate™. It is one of the smallest RF connector line and was developed in the 90s. MMCX is a micro-miniature connector series with a lock-snap mechanism allowing for 360 degrees rotation enabling flexibility.

Adapters, which are also called coaxial adapters, are short, two-sided connectors which are used to join two cables or components which cannot be connected directly. In particular, there are two kinds of adapters. The first one are adapters used to fit connectors of different types. For example an adapter can be used to connect a SMA connector to a BNC one.



Another kind of adapter is the one used to fit connectors of the same type but which cannot be directly joined because of their gender. For example, a very useful adapter is the one which enables to join two Type N connectors, having socket (female) connectors on both sides. In this figure, one of these adapters is shown.

Practical Hints: How to choose the proper connector

- “The gender question”: connector does have a well-defined gender. Usually cables have male connectors at both sides, while RF devices (i.e. transmitters and antennas) have female connectors. Devices as directional couplers and line-through measuring devices may have both male and female connectors. Moreover, connectors have their own threading: it is usually right-hand, but left-hand thread exists as well, especially for WiFi devices due to some US regulations. Be careful about it.
- “The less the best!”: try to minimize the number of connectors/adapters in the RF chain, because each of them introduces some loss (even some dB for each connection!).
- “Buy, don’t build!”: if possible, buy cables that are already terminated with the connectors you need. Soldering connectors is not an easy task, and to do this job properly is almost impossible for small connectors as MCX and MMCX. Even terminating “Foam” cables is not an easy task.
- don’t use BNC for 2.4GHz or higher, use N type connectors (or SMA, SMB, MCX, etc.).
- microwave connectors are precision-made parts, and can be easily damaged by mistreatment. As a general rule, if the connectors have threaded sleeves, you should rotate these to tighten, leaving the rest of the connector (and cable) stationary. If other parts of the connector are twisted while tightening or loosening, damage can easily occur.
- never step over connectors, or drop connectors on the floor when disconnecting cables (this happens more often than what you may imagine, especially when working on a mast over a roof).
- never use tools like pliers to tighten connectors, use your hands. When working outside, remember that metals expand at high temperatures and reduce their size at low temperatures: a very tightened connector in the summer can even break in winter.

Impedance and Impedance Matching

Transmission system behavior differs at low and high frequencies, and the different behaviors are usually described in terms of lumped-constant and distributed-constant systems. Lumped-constant circuits involve components (coils, resistors, capacitors,

etc.) whose physical dimensions are much less than the wavelength of the propagating electromagnetic wave and which can be located at discrete points. When circuit components and connecting wires are of dimensions comparable to a wavelength of the propagating electromagnetic wave, then the circuit components and the wires effectively become distributed constants. We may then think of a line as being composed of a series of small inductors and capacitors, where each coil is the inductance of an extremely small section of wire, and the capacitance is that existing between the same two sections. Each series inductor acts to limit the rate at which current can charge the following shunt capacitor, and in so doing it establishes a very important property of a transmission line, its characteristic impedance. This is abbreviated by convention as Z_0 .

The value of the characteristic impedance is equal to $\sqrt{\frac{L}{C}}$

in a perfect line, one in which the conductors have no resistance and there is no leakage between them. L and C are respectively the inductance and capacitance per unit of length of line. The inductance decreases with increasing conductor diameter, and the capacitance decreases with increasing spacing between the conductors. Hence a line with closely spaced large conductors has a low characteristic impedance, while one with widely spaced thin conductors has a high one. Typical coaxial lines can have characteristic impedance's ranging from 30 Ω to 100 Ω , but most common impedance values for coaxial cables are 50 Ω and 75 Ω . Physical constraints on practical wire diameters and spacing limit Z_0 values to these ranges. The 50 Ω RG-58 cable was developed during World War II to connect antennas which had an impedance of 50 Ω .

A line terminated in a purely resistive load equal to the characteristic line impedance is said to be *matched*. In a matched transmission line, the power is transferred outward from the source until it reaches the load, where it is completely absorbed. Thus with either an infinitely long line or a matched one, the impedance presented to the source of power is the same, regardless of the line length: it is equal to the characteristic impedance of the line. The current in such a line is given by the applied voltage divided by the characteristic impedance, according to Ohm's law. If the terminating resistance R is not equal to Z_0 , then the line is said to be *mismatched*. The more the R differs from Z_0 , the greater the mismatch. The power reaching R is not totally absorbed, as it was when R was equal to Z_0 , because R requires a voltage to current ratio that is different from the one traveling along the line. The result is that R absorbs only part of the power reaching it, the *incident or direct power*. The remainder goes back along the line toward the source, and it is known as the *reflected power*. The greater the mismatch, the larger the percentage of the incident power that is reflected. In the extreme case when R is zero (a short circuit) or infinity (an open circuit), all of the power reaching the end of the line is reflected back toward the source. When there is a mismatch, power is transferred in both directions along the line. The voltage to current ratio must be the same for the reflected power and for the incident one, because this ratio is determined by the Z_0 of the line. The actual voltage at any point along the line is the vector sum of the incident voltage and of the reflected voltage, taking into account the phases of each component. The same is true for the current. The effect of the incident and reflected components on the behavior of the line can be understood by considering two limiting cases: the short-circuited line and the open-circuited line. If the line is short-circuited, the voltage at the end of the line must be zero. Thus the incident voltage must disappear at the short. It can do this only if the reflected voltage is opposite in phase and of the same amplitude. The current, however, does not disappear in the short circuit. The incident current flows

through the short and there is in addition the reflected component in phase with it and of the same amplitude. The reflected voltage and current must have the same amplitudes of the incident ones, because no power is dissipated in the short circuit. Reversing the phase of either the current or voltage reverses the direction of the power flow. If the line is open-circuited, the current must be zero at the end of the line. In this case the reflected current must be opposite in phase with the incident current, and with the same amplitude. The reflected voltage must be in phase with the incident voltage, and must have the same amplitude. When there is a finite value of resistance at the end of the line, only part of the power is reflected. That is, the reflected voltage and current are smaller than the incident ones. The amplitudes of the two components are therefore not equal, but the resultant current and voltage are in phase in R because R is a pure resistance.

The ratio of the reflected voltage at a given point on a transmission line to the incident voltage is called the *voltage reflection coefficient*. The voltage reflection coefficient is also equal to the ratio of the incident and reflected currents.

$$\rho = (E_r/E_i) = (I_r/I_i)$$

Where

- ρ is the reflection coefficient
- E_r is the reflected voltage
- E_i is the incident voltage
- I_r is the reflected current
- I_i is the incident current

The reflection coefficient is determined by the relationship between the line's characteristic impedance and the actual load at the end of the line. In most cases, the load is not entirely resistive. It is a complex impedance, consisting of a resistance in series with a reactance. The reflection coefficient is thus a complex quantity, having both amplitude and phase. It can be designated with the letter ρ or with the letter Γ .

The relationship between R_a , the load resistance, X_a , the load reactance, Z_0 , the line characteristic impedance with real part R_0 and reactive part X_0 and the complex reflection coefficient is given by:

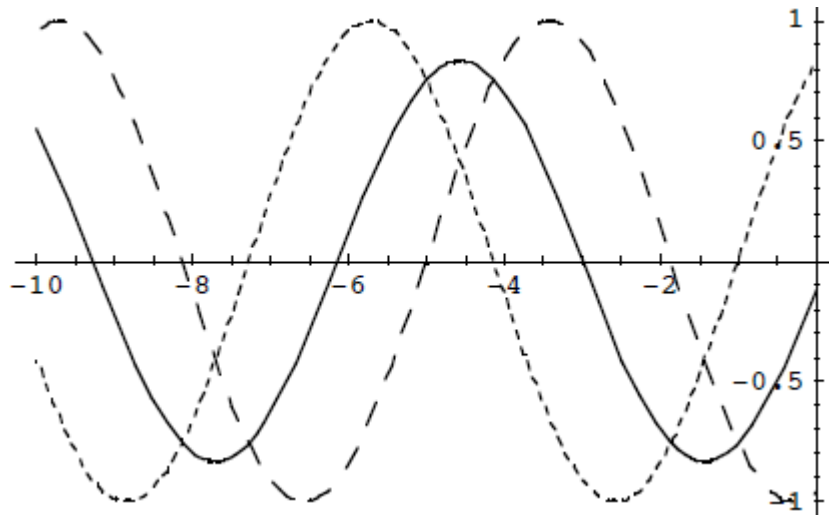
$$\rho = \frac{Z_a - Z_0}{Z_a + Z_0} = \frac{(R_a + jX_a) - (R_0 + jX_0)}{(R_a + jX_a) + (R_0 + jX_0)}$$

For most transmission lines the characteristic impedance is almost completely resistive, meaning that $Z_0 = R_0$ and $X_0 = 0$. The magnitude of the complex reflection coefficient then simplifies to:

$$|\rho| = \sqrt{\frac{(R_a - R_0)^2 + X_a^2}{(R_a + R_0)^2 + X_a^2}}$$

If R_a is equal to R_0 and if X_a is zero, the reflection coefficient is also zero: this is the case of the matched line. On the other hand, if R_a is equal to zero, meaning that the load has no resistive part, then the reflection coefficient is equal to 1 regardless of the value of R_0 . This means that all the forward power is reflected, since the load is completely reactive. As a consequence of reflection, a standing wave may be visualized as an interference between the incident signal E_i at a given frequency, traveling in the forward direction, and the signal E_r , at the same frequency, traveling

in the reverse direction. At the load, the relationship between the amplitudes of E_r and E_i and the phase angle between them are uniquely determined by the load impedance. The phase angle between E_r and E_i , however, will vary along the line as a function of the distance from the load. A wave is created that oscillates in amplitude but never moves laterally. That is why it is called *standing wave*. In the following figure, the E_r , E_i and the standing wave can be seen. The dashed lines are the E_r and the E_i , while the non dashed one represents the standing wave.



At a position 180° from the load ($1/2\lambda$), the voltage and current must have the same values they do at the load. At a position 90° from the load ($1/4\lambda$), the voltage and current must be inverted: if the voltage is lowest and the current is highest at the load, then at 90° from the load the voltage reaches its highest value and the current reaches its lowest value at the same point. Note that the conditions at 90° also exist at 270° , and the ones at 180° are valid at every point multiple of 180° . In a matched line, all of the power that is transferred along the line is absorbed in the load if the load is equal to the characteristic impedance. None of the power is reflected back toward the source. As a result, no standing waves will be developed along the line. The voltage along the line is constant, so the matched line is also said to be flat. The ratio of the maximum voltage, resulting from the interaction of incident and reflected voltages along the line, to the minimum voltage is defined as the *Voltage Standing-Wave Ratio (VSWR)* or simply *Standing-Wave Ratio (SWR)*. The ratio of the maximum current to the minimum current is the same as the VSWR, so either current or voltage can be measured to determine the standing-wave ratio.

$$SWR = \frac{E_{max}}{E_{min}} = \frac{I_{max}}{I_{min}}$$

In the case where the load contains no reactance, the SWR is equal to the ratio between the load resistance R and the characteristic impedance of the line. The standing-wave ratio is an index of many of the properties of a mismatched line. The SWR is related to the magnitude of the complex reflection coefficient by the following equation

$$SWR = \frac{1+|\rho|}{1-|\rho|}$$

where we can see that with $|\rho|=0$ we get $SWR=1$, so we have maximum transmission, and with $|\rho|=1$ we get $SWR=\infty$, so we have no transmission. And the reflection coefficient magnitude may be defined from a measurement of SWR as

$$|\rho| = \frac{SWR - 1}{SWR + 1}$$

We may also express the reflection coefficient in terms of forward and reflected power, quantities that can be easily measured using a directional RF wattmeter. The reflection coefficient may be computed as

$$P = \sqrt{\frac{P_r}{P_f}}$$

where P_f is the power in the forward wave and P_r is the power in the reflected wave. We can use this equation to calculate the SWR from a measurement of the forward and reflected power

$$SWR = \frac{1 + |\rho|}{1 - |\rho|} = \frac{1 + \sqrt{\frac{P_r}{P_f}}}{1 - \sqrt{\frac{P_r}{P_f}}}$$

The relation between the *Return Loss* expresses in dB, which is the amplitude of the reflected wave to the amplitude of the incident wave, and the reflection coefficient is given by:

$$\text{Return Loss (dB)} = -20\log(\rho)$$

The relation between the power ratio and the reflection coefficient is given by:

$$\frac{P_r}{P_f} = 100e^{-2}$$

For example, if we measure a return loss of 15 dB, then we can calculate the reflection coefficient as 0.178 and thus the SWR as 1.43.

Attenuation in Transmission Lines

Every transmission line will have some loss, because of the resistance of the conductors and because power is consumed in the dielectric used for insulating the conductors. Power lost in a transmission line is not directly proportional to the line length, but varies logarithmically with the length. For this reason, line losses are expressed in terms of decibels per unit length, since the decibel is a logarithmic unit. Calculations are very simple because the total loss in a line is found by multiplying the decibel loss per unit length by the total length of the line.

The power lost in a matched line is called *matched-line loss*. It is usually expressed in decibels per 100 feet. It is necessary to specify the frequency for which the loss applies, because the loss varies with frequency. Conductor and dielectric losses increase with frequency, but not in the same way. The relative amount of each type of loss depends also on the construction on the line, so there is no specific relationship between loss and frequency valid for all types of lines. Actual loss values for practical lines can be found in the following table, expressed in dB/100m .

Cable Type	144 MHz	1.2 GHz	2.4 GHz	5.8 GHz
RG-58	20.3	69.2	105.6	169.2
RG-213	9.2	33.1	49.9	93.8
LMR-400	4.9	15.7	22.3	35.4
3/8" LDF	4.3	13.8	19.4	26.6

The power lost in a given line is minimum when the line is terminated in a resistance equal to its characteristic impedance. On non-matched lines there is an additional loss that increases with the increase of the SWR. This is because the effective values of both current and voltage become greater on lines with standing waves. This increase raises the ohmic losses (I^2R) in the conductors and the losses in the dielectric (E^2/R).

Measurements on Transmission Lines

Power and Attenuation Measurements:

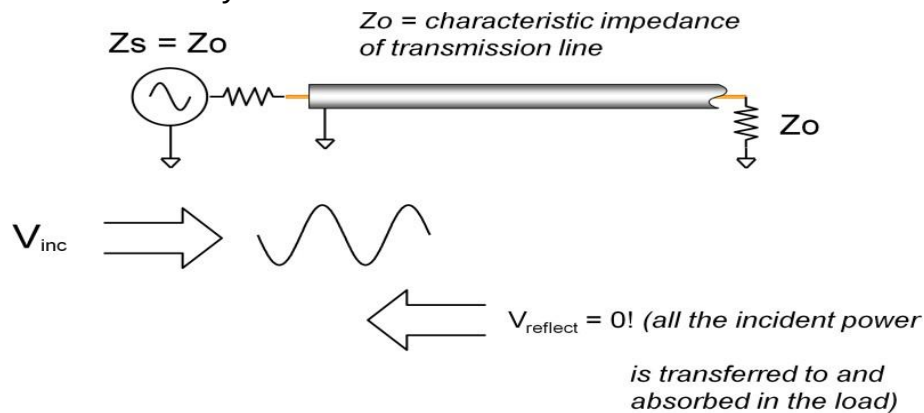
Although a variety of instruments measure power, the most accurate instrument is a *power meter* and a *power sensor*. The sensor is an RF power-to-voltage transducer. The power meter displays the detected voltage as a value of power in log (dBm) or linear (watts) units. Typical power meter instrumentation accuracy will be in the order of hundredths of a dB, while other instruments (i.e., *spectrum analyzers*, *network analyzers*) will have power measurement accuracies in the tenths of dBs or more. One of the main differences between the instruments is that of frequency selective measurements. Frequency selective measurements attempt to determine the power within a specified bandwidth. The traditional Power Meter is not frequency selective in the sense that it measures the average power over the full frequency range of the sensor and will include the power of the carrier as well as any harmonics which may be generated. A Spectrum Analyzer provides a frequency selective measurement since it measures in a particular Resolution Bandwidth. The lack of frequency selectivity is the main reason why Power Meters measure down to around -70 dBm and instruments such as a Spectrum Analyzer can measure much lower than this if narrow resolution bandwidths are used. Average Power provides the average power delivered over several cycles and this is the most common power measurement performed. Average power is defined as the energy transfer rate averaged over many periods of the lowest frequency in the signal. Average power is also defined as the power averaged over a specified time interval. The power meter with sensor only allow to measure the average power, while the Spectrum Analyzer may be used also for more sophisticated power measurements (i.e. peak and pulse power, time-gated power, etc.). We are just interested in average power measurements in the microwave frequency range. Regarding the hardware used in average power measurements, the basic idea behind the power sensor is to convert high frequency power to a DC or low frequency signal that the power meter can then measure and relate to a certain RF power level. There are two types of power sensor: the end-line power sensor which has just one RF input and a connection to the power meter and basically behave as a dummy load, and the line-through power sensor, less common, which has an RF input, and RF output and a connection to the power meter and therefore behaves like a coupler. Often the second type of sensor is integrated in the power meter (line-through power meter).

Transmission line Terminated with LOAD (Z_0)

For maximum transfer of energy into a transmission line from a source or from a transmission line to a load (the next stage of an amplifier, an antenna, etc.), the impedance of the source and load should match the characteristic impedance of the transmission line. In general, then, Z_0 is the target for input and output impedances of devices and networks.

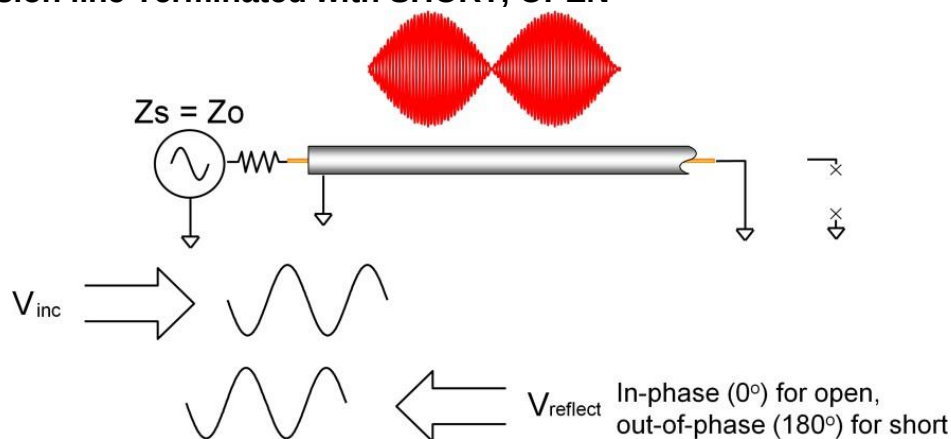
Let's review what happens when transmission lines are terminated in various impedances, starting with a Z_0 load. Since a transmission line terminated in its characteristic impedance results in maximum transfer of power to the load, there is no

reflected signal. This result is the same as if the transmission line was infinitely long. If we were to look at the envelope of the RF signal versus distance along the transmission line, it would be constant (no standing-wave pattern). This is because there is energy flowing in one direction only



For reflection, a transmission line terminated in Z_o behaves like an infinitely long transmission line

Transmission line Terminated with SHORT, OPEN



For reflection, a transmission line terminated in a short or open reflects all power back to source

Next, let's terminate our line in a short circuit. Since purely reactive elements cannot dissipate any power, and there is nowhere else for the energy to go, a reflected wave is launched back down the line toward the source. For Ohm's law to be satisfied (no voltage across the short), this reflected wave must be equal in voltage magnitude to the incident wave, and be 180° out of phase with it. This satisfies the condition that the total voltage must equal zero at the plane of the short circuit. Our reflected and incident voltage (and current) waves will be identical in magnitude but traveling in the opposite direction.

Now let us leave our line open. This time, Ohm's law tells us that the open can support no current. Therefore, our reflected current wave must be 180° out of phase with respect to the incident wave (the voltage wave will be in phase with the incident wave). This guarantees that current at the open will be zero. Again, our reflected and incident current (and voltage) waves will be identical in magnitude, but traveling in the opposite direction. For both the short and open cases, a standing-wave pattern will be set up on the transmission line. The valleys will be at zero and the peaks at twice the incident voltage level.

Planar Transmission Lines

Introduction

Microwave integrated circuit (MIC) techniques represent an extension of integrated circuit technology to microwave frequencies. Since four decades, MIC has increasingly replaced conventional microwave circuits based on coaxial line and waveguide, especially for low power requirements. In MIC technology, all the components are fabricated by the photolithography technology in the planar form. Several forms of planar transmission lines have been proposed for the MIC technology. The planar lines are used not only as interconnects to the components; these are also used to develop passive microwave components and the matching networks. There are several reasons for the wide use of planar transmission lines. First of all, they are broadband, while providing compact dimensions, high reliability, reproducibility and light weight. Foremost, they are generally economical to produce as they are readily adaptable to hybrid and monolithic integrated circuit fabrication technologies at RF and microwave frequencies.

One of the most commonly used transmission lines are the planar types which can be constructed precisely using low-cost printed circuit board materials and processes. A number of these open, multiconductor transmission lines comprise a solid dielectric substrate having one or two layers of metallization, with the signal and ground currents flowing on separate conductors.

Stripline Design

The earliest form of planar transmission lines was straplines which is illustrated in Fig. (1.a) Striplines are essentially modifications of the two wire lines and coaxial lines. It consists of a strip conductor centered between two parallel ground planes with two equal slabs of a dielectric, ferrite, or semiconductor medium separating the center conductor from the ground planes. Usually, the medium is a solid material, but in some applications air is the actual dielectric used. The advantages of striplines are good electromagnetic shielding and low attenuation losses, which make them suitable for high-quality factor (Q) and low-interference applications. Transverse electric and magnetic (TEM) waves propagate within the stripline. Such waves have electric and magnetic components in a plane transverse to the direction of propagation. However, Striplines require strong symmetry and thereby present difficulties in the design of many circuit functions. Also, the tuning of circuits becomes difficult, because it requires destruction of the symmetry to access the center conductor. Any vertical asymmetry in the stripline structure could couple to waveguide-type modes bounded by the ground planes and the side walls. Also, with few exceptions of circuit configuration, the stripline structure is not convenient for incorporating chip elements and associated bias circuitry.

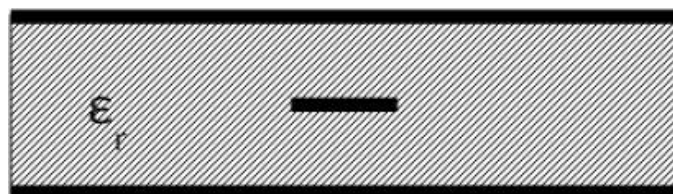


Fig. (1.a): Strip line.

Microstrip Line Design

The microstrip line is transmission line geometry with a single conductor trace on one side of a dielectric substrate and a single ground plane on the other side is shown in Fig.1.b Since it is an open structure, microstrip line has a major fabrication advantage over the stripline. It also features ease of interconnection and adjustments.

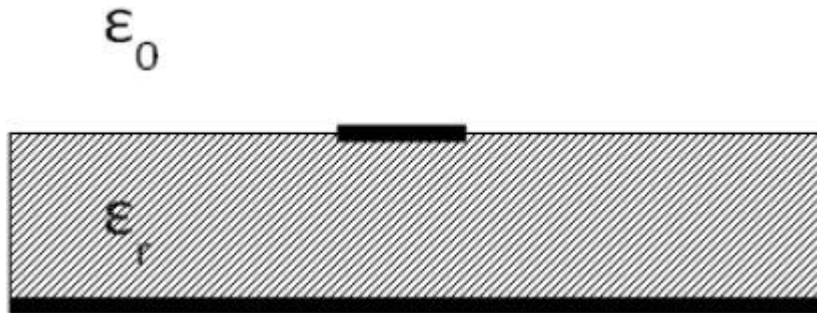


Fig.1.b A microstrip line

In the microstrip line, the electromagnetic fields exist partly in the air above the dielectric substrate and partly within the substrate itself. For most practical purposes, microstrip can be treated as a TEM transmission line with an effective relative permittivity (ϵ_{eff}) that is a weighted average between air and the substrate material. But, the actual propagation of electromagnetic waves in microstrip is not purely TEM due to the combination of an open air space and a dielectric medium. For microwave device applications, microstrip generally offers the smallest sizes and the easiest fabrication.

In general, planar transmission lines consist of strip metallic conductors, usually produced by some photographic process, on a non-conducting substrate. Typical substrate materials are slabs of dielectric, ferrite, or high resistivity semiconductors. In most cases, there are metal ground planes that can either be printed on the same substrate or be a part of the metal housing of MIC. This allows the characteristic impedance (Z_0) of the line to be controlled by defining the dimensions in a single plane. It is to be noted in Fig. (1.c) that the substrate materials with permittivity ϵ_r are denoted by gray areas, and conductors and ground planes by bold lines. The region with free permittivity ϵ_0 is free space or air.

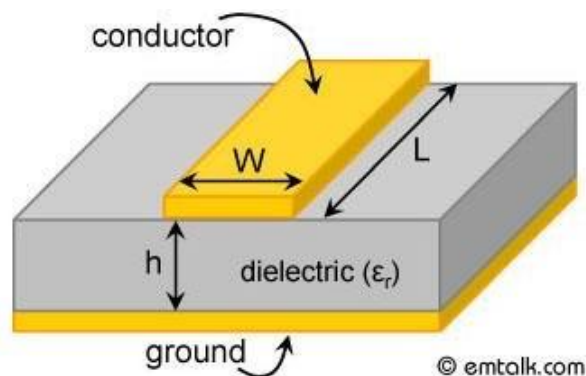


Fig.1.c A microstrip line

Characteristics Impedance of microstrip line is given by

$$\text{If } \left(\frac{W}{H}\right) < 1: \epsilon_{eff} = \frac{\epsilon_R + 1}{2} + \frac{\epsilon_R - 1}{2} \left[\frac{1}{\sqrt{1 + 12\left(\frac{H}{W}\right)}} + 0.04 \left(1 - \left(\frac{W}{H}\right)\right)^2 \right]$$

$$Z_0 = \frac{60}{\sqrt{\epsilon_{eff}}} \ln \left(8 \left(\frac{H}{W} \right) + 0.25 \left(\frac{W}{H} \right) \right)$$

$$\text{If } \left(\frac{W}{H}\right) > 1: \epsilon_{eff} = \frac{\epsilon_R + 1}{2} + \left[\frac{\epsilon_R - 1}{2\sqrt{1 + 12\left(\frac{H}{W}\right)}} \right]$$

$$Z_0 = \frac{120\pi}{\sqrt{\epsilon_{eff}} \left[\frac{W}{H} + 1.393 + \frac{2}{3} \ln \left(\frac{W}{H} + 1.444 \right) \right]}$$

Where ϵ_r the Relative Dielectric Constant
 W is the Width of track
 t is the Thickness of track (1 oz/ft² = 0.035 mm)
 H is the Thickness of dielectric
 Z₀ is Characteristics Impedance

Applications of Planar Transmission Lines

- Filters
- Hybrids
- High-Q resonators

Coplanar Waveguide (CPW) Design

Coplanar Waveguide (CPW) is an alternative to Microstrip and Stripline that place both, the signal and ground currents on the same layer

- The conductors formed a center strip separated by a narrow gap from two ground planes on either side. The dimensions of the center strip, the gap, the thickness and permittivity of the dielectric substrate determined the effective dielectric constant, characteristic impedance and the attenuation of the line.
- The gap in the coplanar waveguide is usually very small and supports electric fields primarily concentrated in the dielectric. With little fringing field in the air space, the coplanar waveguide exhibits low dispersion. In order to concentrate the fields in the substrate area and to minimize radiation, the dielectric substrate thickness is usually set equal to about twice the gap width.

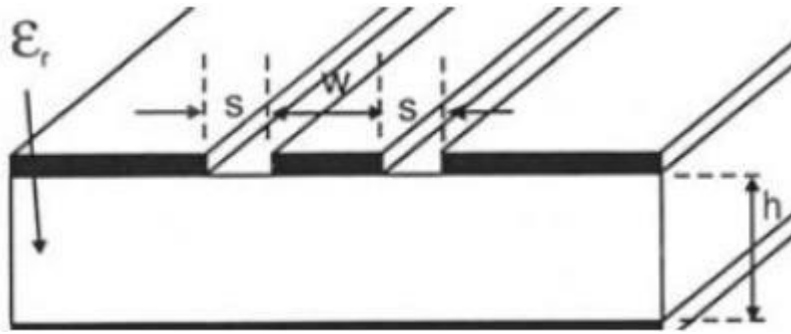


Fig.1.d CPW

The coplanar waveguide (CPW) structure consists of a center strip width with two parallel ground planes equidistant from it on either side, as shown in Fig. (1.d). The center conductor and ground planes are located in one plane on the substrate surface.

Table-(1.1): Characteristics and Applications of the Various Planar Transmission Lines

Transmission Line	Q - Factor	Radiation	Dispersive	Impedance Range (Ω)	Chip Mounting	Applications
Stripline	400	Low	None	35-250	Poor	Blocking filters
Microstrip Line	250 (dielectric substrate) 100-150 (Si, GaAs substrate)	Low (for high ϵ_r) High (for low ϵ_r)	Low	20-120	Difficult for shunt; easy for series	- Filters - Hybrids - High-Q resonators
CPW	150	Medium	Low	20-250	Easy for series and shunt	- Filters - Hybrids - High-Q resonators

Smith Chart

The Smith chart, invented by Phillip H. Smith (1905–1987) is a graphical aid or nomogram designed for electrical and electronics engineers specializing in radio frequency (RF) engineering to assist in solving problems with transmission lines and matching circuits. The Smith chart can be used to simultaneously display multiple parameters including impedances, admittances, reflection coefficients, scattering parameters, noise figure circles, constant gain contours and regions for unconditional stability, including mechanical vibrations analysis. The Smith chart is most frequently used at or within the unity radius region. However, the remainder is still mathematically relevant, being used, for example, in oscillator design and stability analysis While the use of paper Smith charts for solving the complex mathematics involved in matching

problems has been largely replaced by software based methods, the Smith charts display is still the preferred method of displaying how RF parameters behave at one or more frequencies, an alternative to using tabular information. Thus most RF circuit analysis software includes a Smith chart option for the display of results and all but the simplest impedance measuring instruments can display measured results on a Smith chart display.

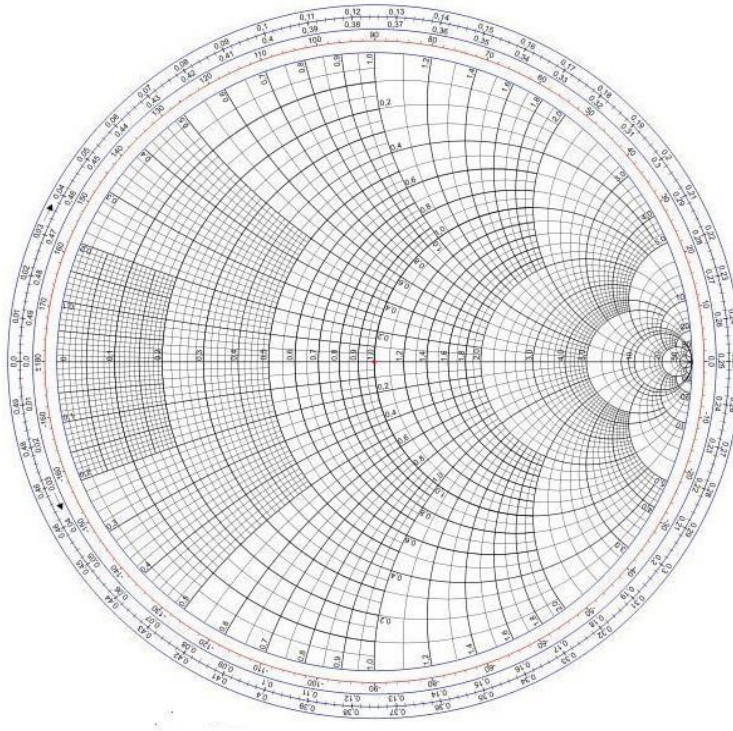


Fig.1 basics Smith chart

The Smith chart is plotted on the complex reflection coefficient plane in two dimensions and is scaled in normalised impedance (the most common), normalised admittance or both, using different colours to distinguish between them. These are often known as the Z, Y and YZ Smith charts respectively. Normalised scaling allows the Smith chart to be used for problems involving any characteristic or system impedance which is represented by the center point of the chart. The most commonly used normalization impedance is 50 ohms. Once an answer is obtained through the graphical constructions described below, it is straightforward to convert between normalised impedance (or normalised admittance) and the corresponding unnormalized value by multiplying by the characteristic impedance (admittance). Reflection coefficients can be read directly from the chart as they are unit less parameters. The Smith chart has circumferential scaling in wavelengths and degrees. The wavelengths scale is used in distributed component problems and represents the distance measured along the transmission line connected between the generator or source and the load to the point under consideration. The degree's scale represents the angle of the voltage reflection coefficient at that point. The Smith chart may also be used for lumped element matching and analysis problems. Use of the Smith chart and the interpretation of the results obtained using it requires a good understanding of AC circuit theory and transmission line theory, both of which are pre-requisites for RF engineers.

As impedances and admittances change with frequency, problems using the Smith chart can only be solved manually using one frequency at a time, the result being represented by a point. This is often adequate for narrow band applications (typically up

to about 5% to 10% bandwidth) but for wider bandwidths it is usually necessary to apply Smith chart techniques at more than one frequency across the operating frequency band. Provided the frequencies are sufficiently close, the resulting Smith chart points may be joined by straight lines to create a locus.

A locus of points on a Smith chart covering a range of frequencies can be used to visually represent:

- how capacitive or how inductive a load is across the frequency range
- how difficult matching is likely to be at various frequencies
- how well matched a particular component is.

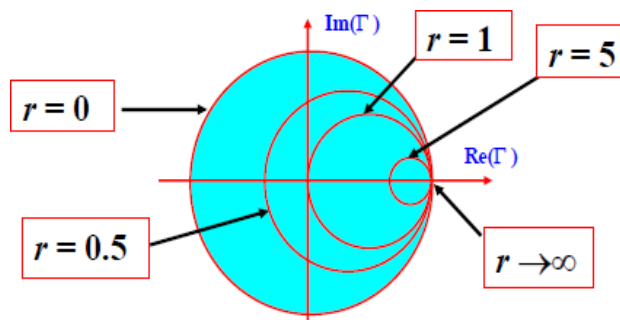
The accuracy of the Smith chart is reduced for problems involving a large locus of impedances or admittances, although the scaling can be magnified for individual areas to accommodate these.

The Smith Chart is a fantastic tool for visualizing the impedance of a transmission line and antenna system as a function of frequency. Smith Charts can be used to increase understanding of transmission lines and how they behave from an impedance viewpoint. Smith Charts are also extremely helpful for impedance matching, as we will see. The Smith Chart is used to display an actual (physical) antenna's impedance when measured on a Vector Network Analyzer (VNA).

The result for the real part indicates that on the complex plane with coordinates ($\text{Re}(\Gamma)$, $\text{Im}(\Gamma)$) all the possible impedances with a given normalized resistance r are found on a circle with

$$\text{Center} = \left\{ \frac{r}{1+r}, 0 \right\} \quad \text{Radius} = \frac{1}{1+r}$$

As the normalized resistance r varies from 0 to ∞ , we obtain a family of circles completely contained inside the domain of the reflection coefficient $|\Gamma| \leq 1$



Return Loss to VSWR Conversion Table

Return Loss (dB)	VSWR	Reflection Coefficient, Γ	Mismatch Loss (dB)	Reflected Power (%)	Through Power (%)
1	17.39	0.891	6.868	79.43	20.57
2	8.72	0.794	4.329	63.10	36.90
3	5.85	0.708	3.021	50.12	49.88
4	4.42	0.631	2.205	39.81	60.19
5	3.57	0.562	1.651	31.62	68.38
6	3.01	0.501	1.256	25.12	74.88
7	2.61	0.447	0.967	19.95	80.05
8	2.32	0.398	0.749	15.85	84.15
9	2.10	0.355	0.584	12.59	87.41
10	1.92	0.316	0.458	10.00	90.00
11	1.78	0.282	0.359	7.94	92.06
12	1.67	0.251	0.283	6.31	93.69
13	1.58	0.224	0.223	5.01	94.99
14	1.50	0.200	0.176	3.98	96.02
15	1.43	0.178	0.140	3.16	96.84
16	1.38	0.158	0.110	2.51	97.49
17	1.33	0.141	0.088	2.00	98.00
18	1.29	0.126	0.069	1.58	98.42
19	1.25	0.112	0.055	1.26	98.74
20	1.22	0.100	0.044	1.00	99.00
21	1.20	0.089	0.035	0.79	99.21
22	1.17	0.079	0.027	0.63	99.37
23	1.15	0.071	0.022	0.50	99.50
24	1.13	0.063	0.017	0.40	99.60
25	1.12	0.056	0.014	0.32	99.68
26	1.11	0.050	0.011	0.25	99.75
27	1.09	0.045	0.009	0.20	99.80
28	1.08	0.040	0.007	0.16	99.84
29	1.07	0.035	0.005	0.13	99.87
30	1.07	0.032	0.004	0.10	99.90
31	1.06	0.028	0.003	0.08	99.92
32	1.05	0.025	0.003	0.06	99.94
33	1.05	0.022	0.002	0.05	99.95
34	1.04	0.020	0.002	0.04	99.96
35	1.04	0.018	0.001	0.03	99.97
36	1.03	0.016	0.001	0.03	99.97
37	1.03	0.014	0.001	0.02	99.98
38	1.03	0.013	0.001	0.02	99.98
39	1.02	0.011	0.001	0.01	99.99
40	1.02	0.010	0.000	0.01	99.99

$$\Gamma = 10^{(-\text{Return Loss}/20)}$$

$$\text{VSWR} = [1 + 10^{(-\text{Return loss}/20)}] / [1 - 10^{(-\text{Return loss}/20)}]$$

$$\text{VSWR} = (1 + |\Gamma|) / (1 - |\Gamma|)$$

$$\text{Mismatch Loss (dB)} = 10 \log(1 - \Gamma^2)$$

$$\text{Reflected Power (\%)} = 100 * \Gamma^2$$

$$\text{Return Loss (dB)} = -20 \log |\Gamma|$$

$$\text{Return Loss (dB)} = -20 \log [(VSWR-1)/(VSWR+1)]$$

$$\Gamma = (VSWR-1)/(VSWR+1)$$

$$\text{Through Power (\%)} = 100 (1 - \Gamma^2)$$

EXPERIMENT NO.1

EXPERIMENT NO:1

AIM:-

To understand the purpose and operational principle of the standing wave using maxima & minima.

OBJECTIVES:-

To explore the properties of TRANSMISSION LINE

- (a) analysis of minima and maxima creation at various frequency
- (b) observation of standing wave pattern at various frequency

Outcome of this Experiment:

Concept of transmission

- Understanding the transmission line design topology

EQUIPMENTS:-

List of equipment's and accessories needed for performing the experiments

DUT, Equipments and Accessories	Model Name	Quantity
Micro strip transmission line	Transmission Line JIG	1
RFL-MMTLT Control Unit	RFL-MMTLT Control Unit	1
Cables	SMA Cable – 1 meter	1
	SMA Cable -2meter	1
	9 Pin D Type Male to Female Cable	1
	9 Pin D Type Male to DIN Connector Cable	1
Termination	Load	1
Software	Transmission Line Software on PC	1

SET UP & PROCEDURE:-

PC BASED CONTROL

Hardware Connections:

1. Use “9 Pin D Type Male to Female Cable” to connect the “stepper motor” located on Control Unit and 9 Pin D type Male Terminal Located on Transmission Line Jig.
2. Use “ 9 Pin D Type Male to DIN (female) Connector Cable “ to connect between Limit Switch (9 Pin D type Female Connector) on back side of the Control Unit and 9 Pin DIN (Male) Type Connector on Transmission Line Jig.

3. Connect the RF output of the Motorized Microstrip Transmission Line Trainer (**RF OUT**) to the port 1 (Fig.1) of the transmission Line JIG using 1 meter SMA(m) to SMA (m) Cable (i.e. Signal generated by RFL-MMTLT RF Signal Generator is applied as input to the transmission line)

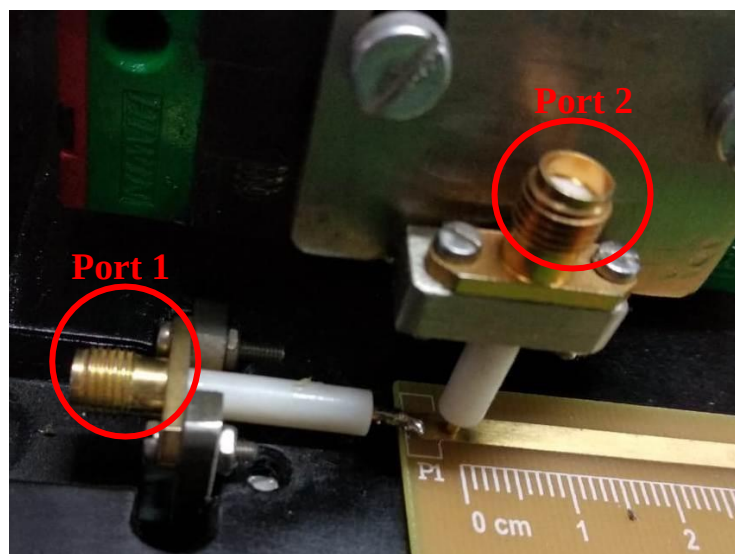


Fig.1

4. Connect the RF DETECTOR (RF IN) on the Control Unit to Port 2 of the transmission line JIG as shown in (Fig.1) using 2 meter SMA (m) to SMA (m) Cable .

5. Connect Load at Port 3 of transmission line assembly (Fig.2)

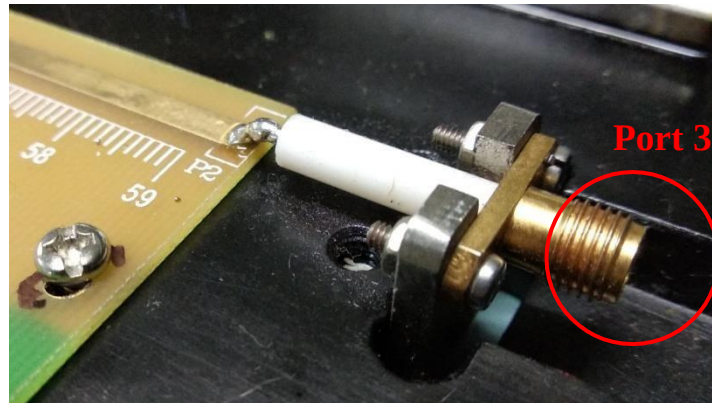


Fig.2

6. Turn ON the RFL-MMTLT System, Apply Power using Power Cable and switch ON the Trainer Kit using Power ON switch available on the rare side of the Control Unit

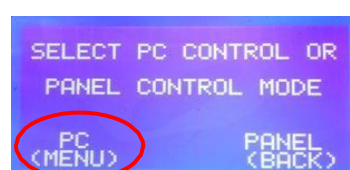
Display will show following messages



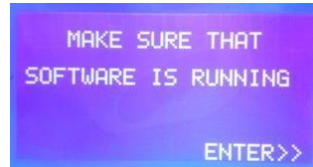
7. Connect the USB cable from the Rare Side of the Control Unit to the USB Port of the Personal Computer

8. Press **MENU** button on the Control unit

9. To select the PC Control Mode, Press MENU key.



10. Make sure that Software is running and press **ENTER**.



11. Display will show Software Control Mode Selected throughout the experiment.



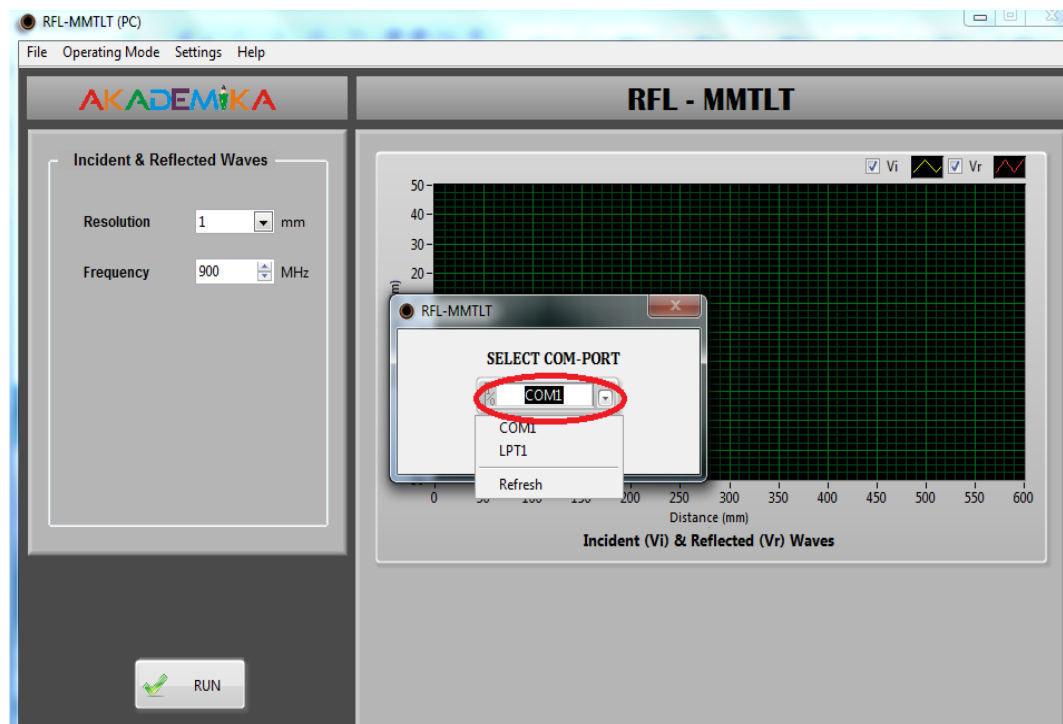
12. Double click the RFL- MMTLT icon to run the Software in PC



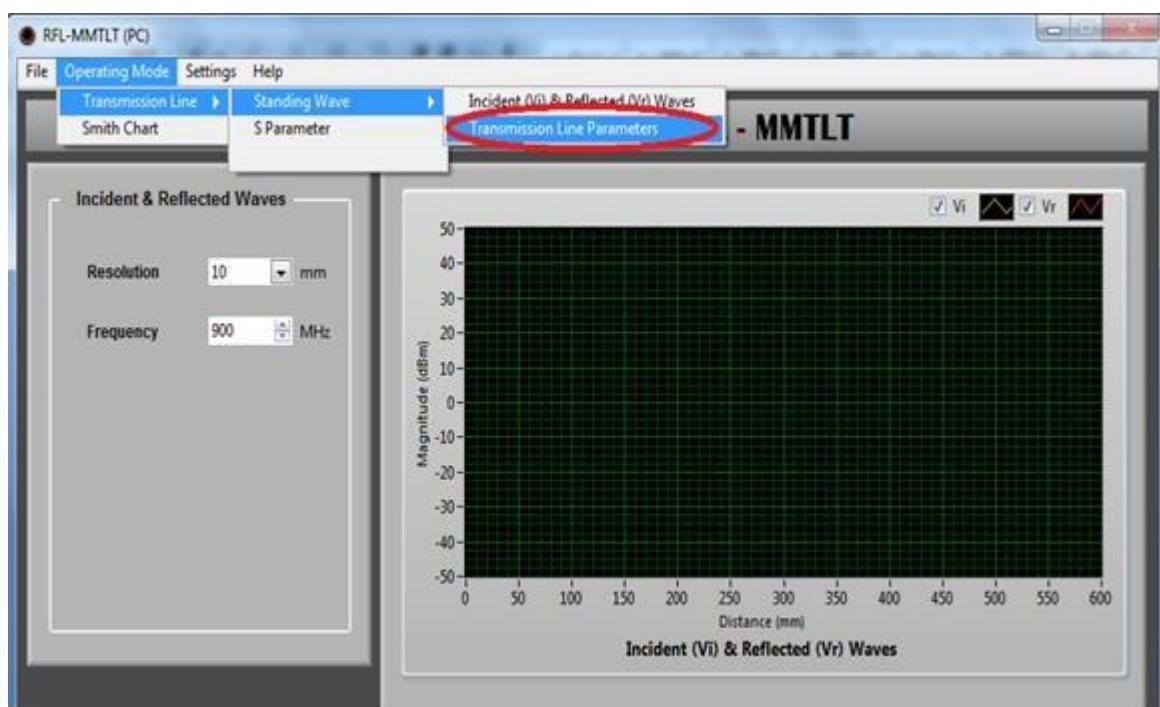
Select **File>Control Mode>PC** to choose PC control Mode



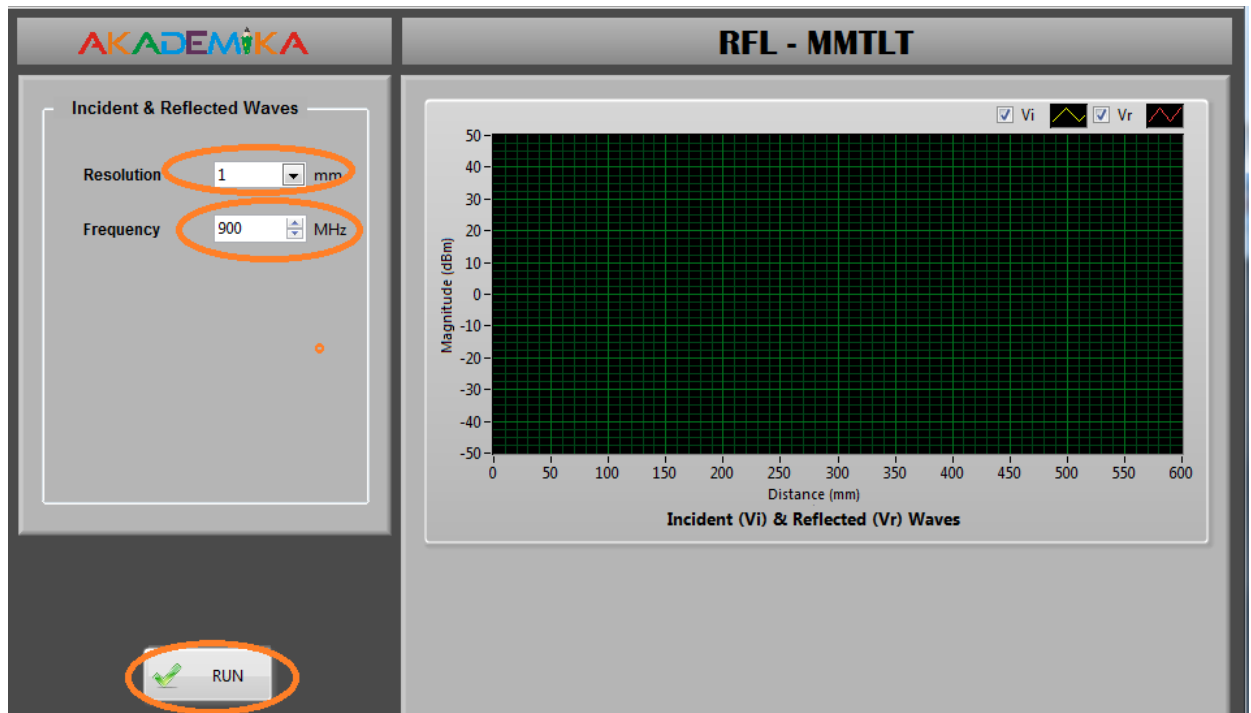
Select **COM PORT**: - Below picture shows the selection of com port. Click refresh button to update the COM port. (Need to Install the Drivers for the FT232 given along with the software)



Select transmission line parameter as shown below



Select resolution and frequency **1mm & 900MHz** respectively

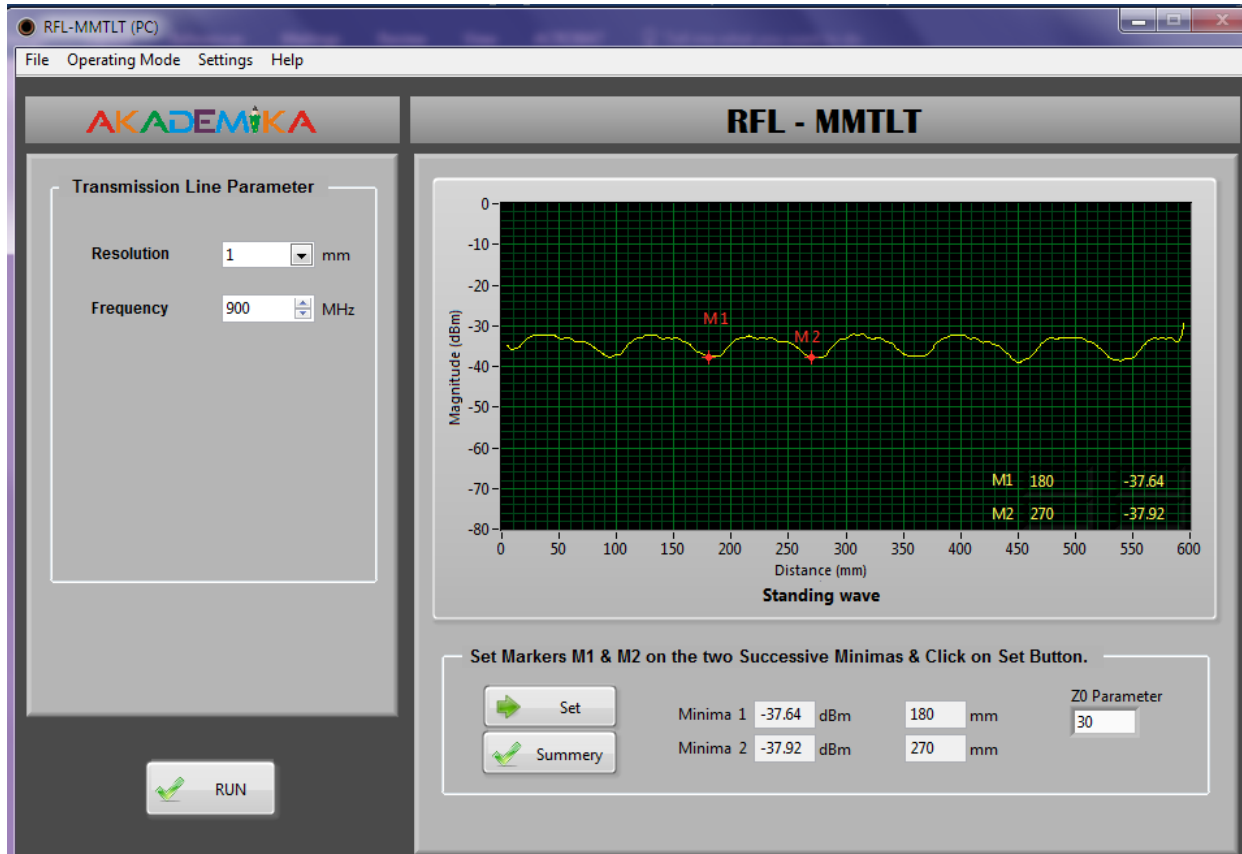


Click on RUN button to start experiment

- When we press RUN button motor starts running and signal sensing probe at port2 starts moving from 0cm of transmission line towards 59cm (Total 600mm). After completion of this distance moving probe returns back to 0cm position.

Note: During start of experiment, if moving probe position is not at 0cm, you can still RUN the experiment. It automatically comes to 0cm i.e. home position and then starts the experiment.

The sensor probe takes some minutes to complete its full rotation from generator to load the graph will display on screen. Save the graph on computer.



- To save the image of wave go to **file>save as>Image File**
- To save the excel data of wave go to **file>save as>CSV File**
- Now repeat the same procedure to take readings for frequency 800MHz and 1000MHz
- Save multiple frequency waves in CSV for to particular location and the compare all the graphs
- To compare the graphs, go to **Settings>plot comparison**
- Browse the graphs you want to compare and observe the waves on software.

OBSERVATION TABLE FOR PC MODE

Total Number of maxima observed for 900 MHz				Total Number of minima observed for 900 MHz			
Frequency 900 MHz	1 st minima	1 st maxima	2 nd minima	2 nd maxima	3 rd minima	3 rd maxima	4 th minima
Power level in dBm or dBuV							
Location of movable port (cm)							

Total Number of maxima observed for 1200 MHz				Total Number of minima observed for 800 MHz			
Frequency 1200 MHz	1 st minima	1 st maxima	2 nd minima	2 nd maxima	3 rd minima	3 rd maxima	4 th minima
Power level in dBm or dBuV							
Location of movable port (cm)							

Total Number of maxima observed for 500 MHz				Total Number of minima observed for 1000 MHz			
Frequency 500 MHz	1 st minima	1 st maxima	2 nd minima	2 nd maxima	3 rd minima	3 rd maxima	4 th minima
Power level in dBm or dBuV							
Location of movable port (cm)							

Conclusion: -

1. Effect of frequency on number of minima and maxima.....
2. Effect of frequency on two successive Minima or maxima.....
3. Effect of frequency on magnitude of 2nd minima & 2nd maxima.....

EXPERIMENT NO.2

EXPERIMENT NO: 2

NAME:-

Experimentation on OPEN, SHORT, LOAD characterization

OBJECTIVE:-

To comprehend the effect of the various load condition on microstrip transmission line

AIM:-

To explore the properties of the TRANSMISSION LINE module

- (a) Observation & characterization for given LOADS.
- (b) Measurement of minima & maxima.

EQUIPMENTS:-

List of equipment's needed for performing the experiments

DUT, Equipment's and Accessories	Model Name	Quantity
Micro strip transmission line	Transmission line JIG	1
RFL-MMTLT Control Unit	RFL-MMTL Control Unit	1
Cables	SMA Cable – 1 meter SMA Cable -2meter 9 Pin D Type Male to Female Cable 9 Pin D Type Male to DIN Connector Cable	1 1 1 1
Termination	Open (O) Short (S) Load (L)	1 1 1
Software	Transmission Line Software on PC	1

SET UP & PROCEDURE:-

LOAD Condition :-

Hardware Connections:

1. Use "9 Pin D Type Male to Female Cable" to connect the "stepper motor" located on Control Unit and 9 Pin D type Male Terminal Located on Transmission Line Jig.
2. Use " 9 Pin D Type Male to DIN (female) Connector Cable " to connect between Limit Switch (9 Pin D type Female Connector) on back side of the Control Unit and 9 Pin DIN (Male) Type Connector on Transmission Line Jig.

3. Connect the RF output of the Motorized Microstrip Transmission Line Trainer (**RF OUT**) to the port 1 (Fig.1) of the transmission Line JIG using 1 meter SMA(m) to SMA (m) Cable (i.e. Signal generated by RFL-MMTLT RF Signal Generator is applied as input to the transmission line)
4. Connect the RF DETECTOR (RF IN) on the Control Unit to Port 2 of the transmission line JIG using 2 meter SMA (m) to SMA (m) Cable .
5. Connect LOAD at Port 3 of transmission Line JIG.
6. Turn ON the RFL-MMTLT System, Apply Power using Power Cable and switch ON the Trainer Kit using Power ON switch available on the rare side of the Control Unit



7. Connect the USB cable from the Rare Side of the Control Unit to the USB Port of the Personal Computer
8. Press **MENU** button on the Control unit.
9. Select PC Control Mode. Press MENU key for PC control mode.
10. Make sure that Software is running and press **ENTER**.
11. Display will show Software Control Mode Selected throughout the experiment..



12. Double click the RFL- MMTLT icon to run the Transmission Line Software in PC
13. Select the **COM PORT**: Click refresh button to update the COM port. (Need to Install the Drivers for the FT232 given along with the software)
14. Select transmission line parameter on the software as done in the previous experiment.
15. Select resolution and frequency 1mm & 900MHz respectively

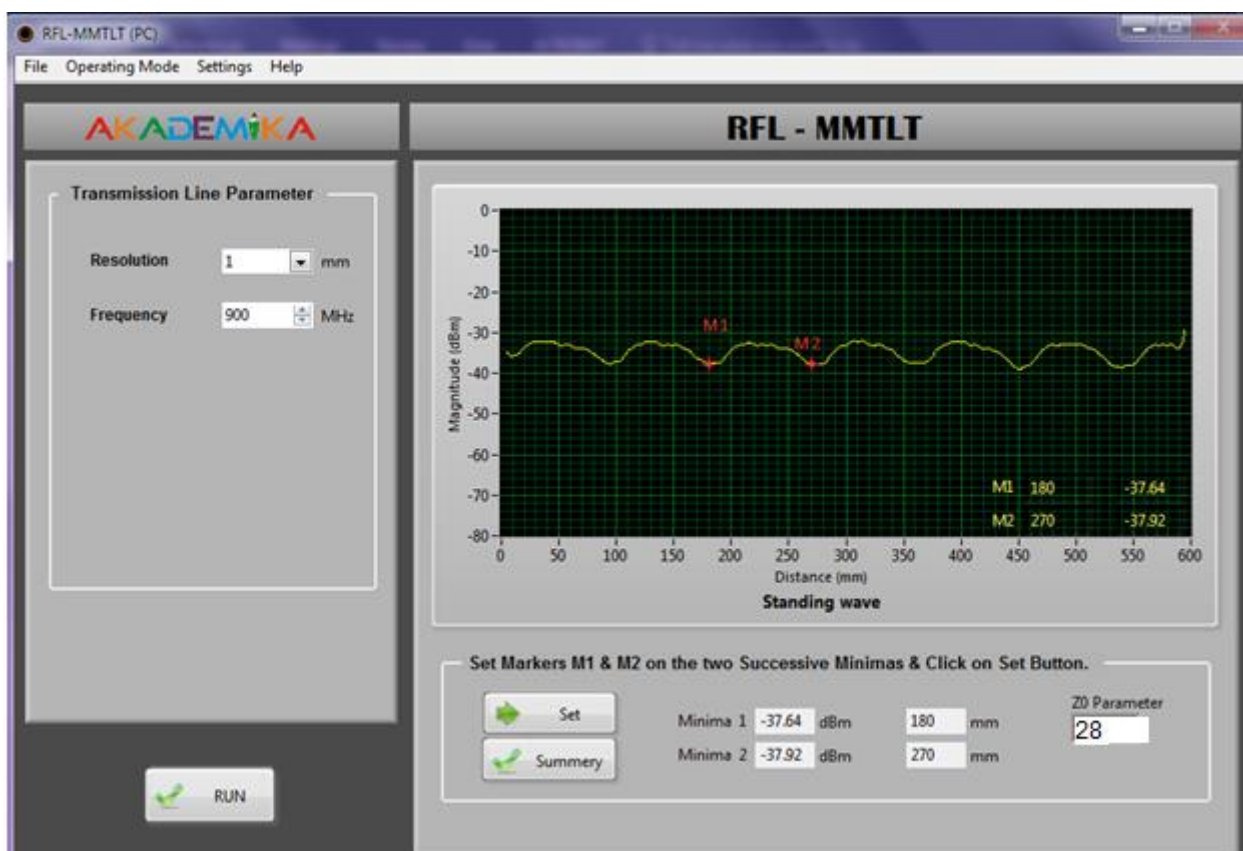
Click on RUN button to start experiment

- When we press RUN button motor starts running and signal sensing probe at port2 starts moving from 0cm of transmission line towards 59cm (Total 600mm). After completion of this distance moving probe returns back to 0cm position.

Note: During start of experiment if moving probe position is not at 0cm, you can still **RUN** the experiment. It automatically comes to 0cm i.e. home position and then starts the experiment.

The sensor probe takes some minutes to complete its full rotation from generator to load. The graph will display on screen. Save the graph on computer.

Go to **Settings>Plot settings** to initiate the markers



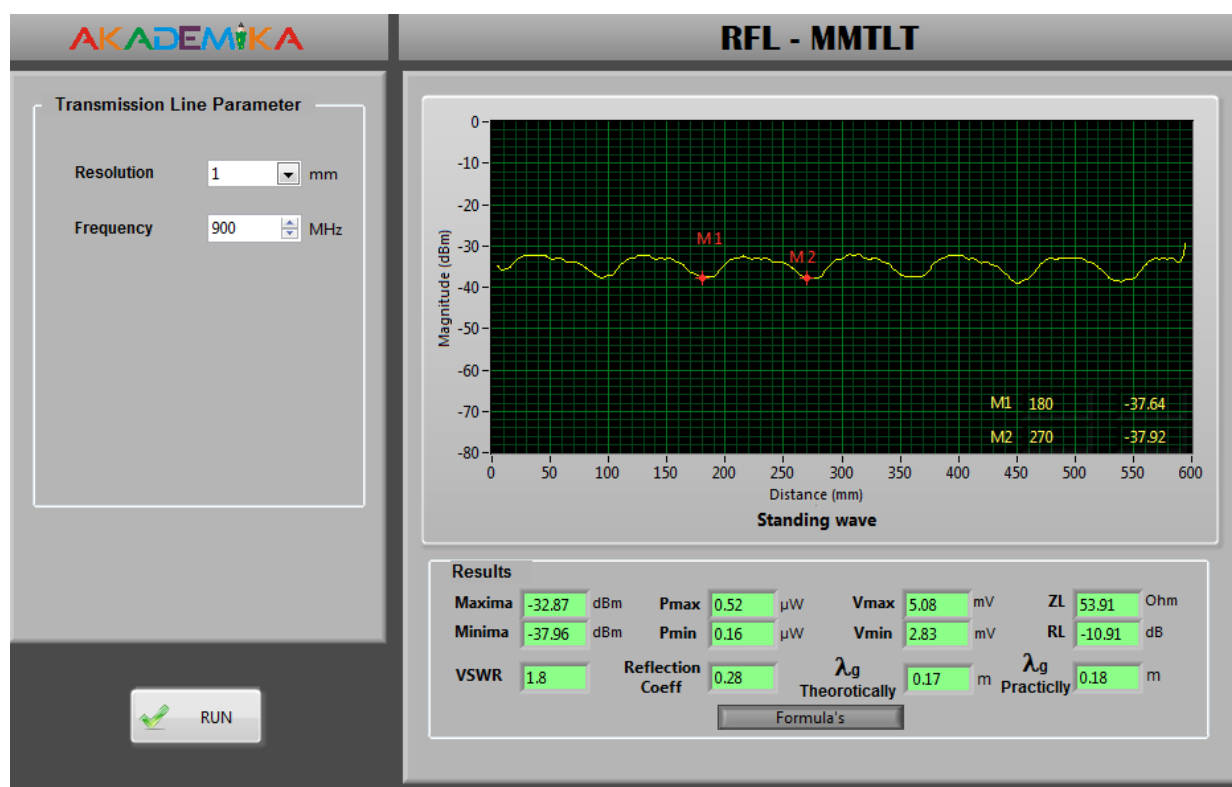
Now on graph you can move the markers to find the Distance and Magnitude of particular Minima and maxima.

Keep Markers on two successive minima's and Insert **Zo parameter is set to 28**.

Refer Annexure 1

(Note: This is impedance ($Z_0=28$) of transmission line with LOAD termination which is measured on VNA at 900MHz frequency)

Press **Summary** to get all the transmission Line Parameters for your selected Load Condition.



- To save the image of wave go to **file>save as>Image File**
- To save the excel data of wave go to **file>save as>CSV File**

OBSERVATION TABLE FOR LOAD Condition

Frequency 900 MHz	1 st minima dBm	1 st maxima dBm	2 nd minima dBm	2 nd Maxima dbm	3 rd minima dBm	3 rd maxima dBm
Location of movable port (cm)						

STEP1

Consider any adjacent minima and maxima

- $P_{\max} = 2^{\text{nd}}$ maxima (dbm)
- $P_{\min} = 2^{\text{nd}}$ minima (dbm)

(Consider any adjacent combination of minima and maxima.eg; - 2minima & 2maxima or 3rd minima & 3rd maxima)

Step2

Convert to dbm to mW conversion

$$P_{\text{out}} (\text{mW}) = 1000 \times 10^{(\text{dBm} / 10)}$$

- $P_{\max} (\text{mW}) = 1000 \times 10^{(2^{\text{nd}} \text{ maxima} / 10)}$
- $P_{\min} (\text{mW}) = 1000 \times 10^{(2^{\text{nd}} \text{ minima} / 10)}$

STEP3

- **Calculation of Vmax**

- $V_{\max} = \sqrt{P_{\max} * Z_0}$

- **Calculation of Vmin**

- $V_{\min} = \sqrt{P_{\min} * Z_0}$

- **Calculation of VSWR**

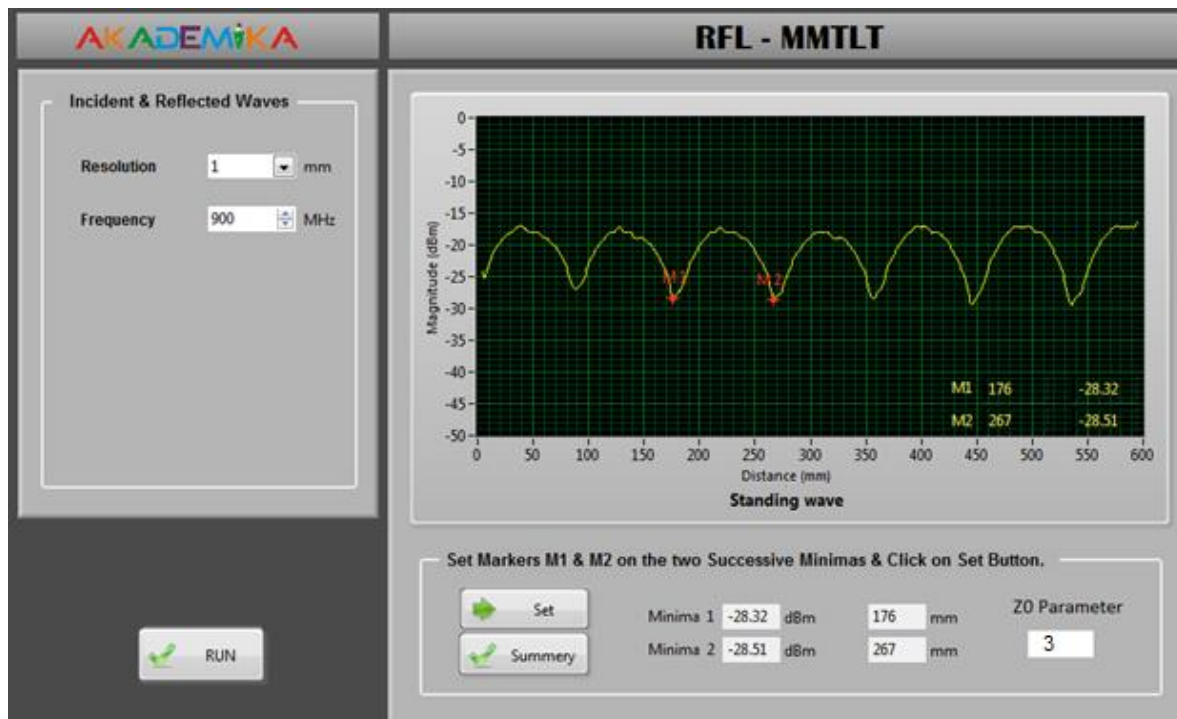
- $VSWR = \frac{V_{\max}}{V_{\min}}$

SHORT Condition:

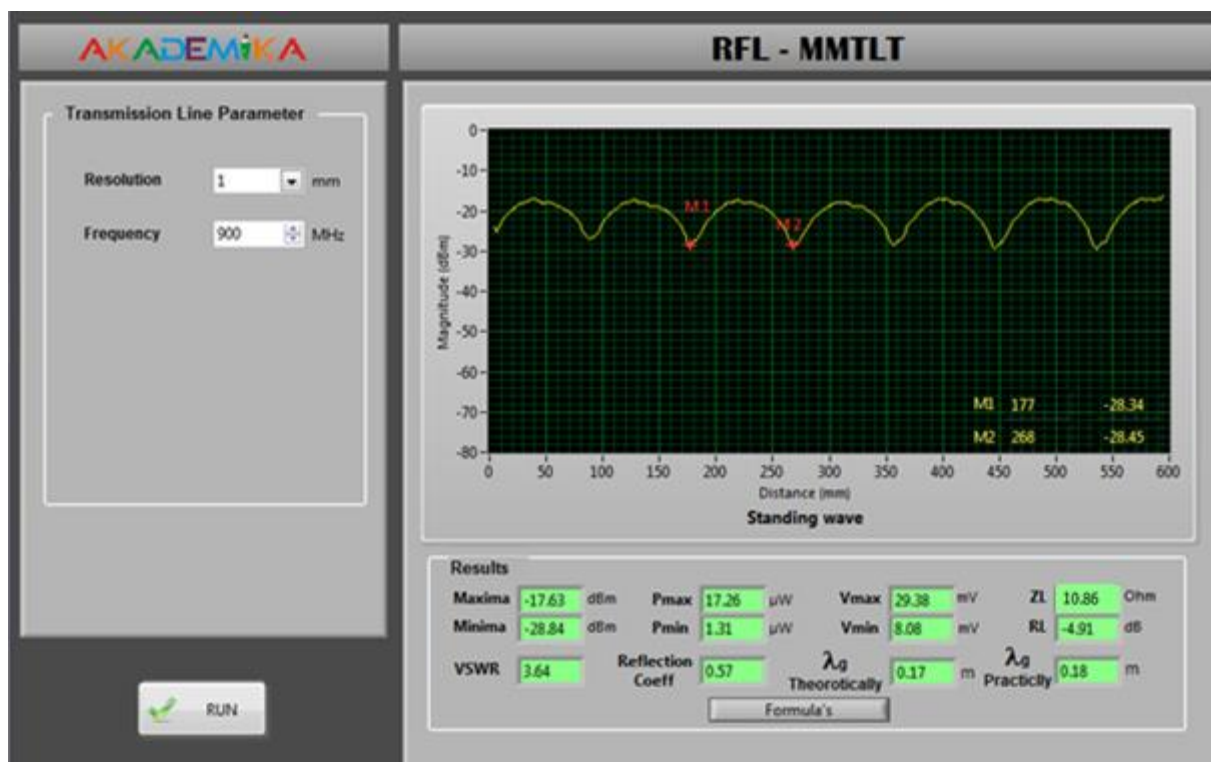
Follow the same procedure and replace the Port 3 of the Transmission Line JIG with SHORT termination condition and get the results

For Short Category Insert Z_0 parameter as 3 (Refer Annexure 1)

(Note: This is impedance ($Z_0=3$) of transmission line with SHORT termination which is measured on VNA at 900MHz frequency)



Press **Summary** to get all the transmission Line Parameters for your selected SHORT termination condition



OBSERVATION TABLE FOR SHORT Condition

Frequency 900 MHz	1 st minima dBm	1 st maxima dBm	2 nd minima dBm	2 nd Maxima dbm	3 rd minima dBm	3 rd maxima dBm
Location of movable port (cm)						

STEP1

Consider any two adjacent minima and maxima

- $P_{\max} = 2^{\text{nd}} \text{ maxima (dbm)}$
- $P_{\min} = 2^{\text{nd}} \text{ minima (dbm)}$

(Consider any adjacent combination of minima and maxima.eg; - 2minima & 2maxima or 3rd minima & 3rd maxima)

Step2

Convert to dbm to mW conversion

$$P_{\text{out}} (\text{mW}) = 1000 \times 10^{(\text{dBm} / 10)}$$

- $P_{\max} (\text{mW}) = 1000 \times 10^{(2^{\text{nd}} \text{ maxima} / 10)}$
- $P_{\min} (\text{mW}) = 1000 \times 10^{(2^{\text{nd}} \text{ minima} / 10)}$

STEP3

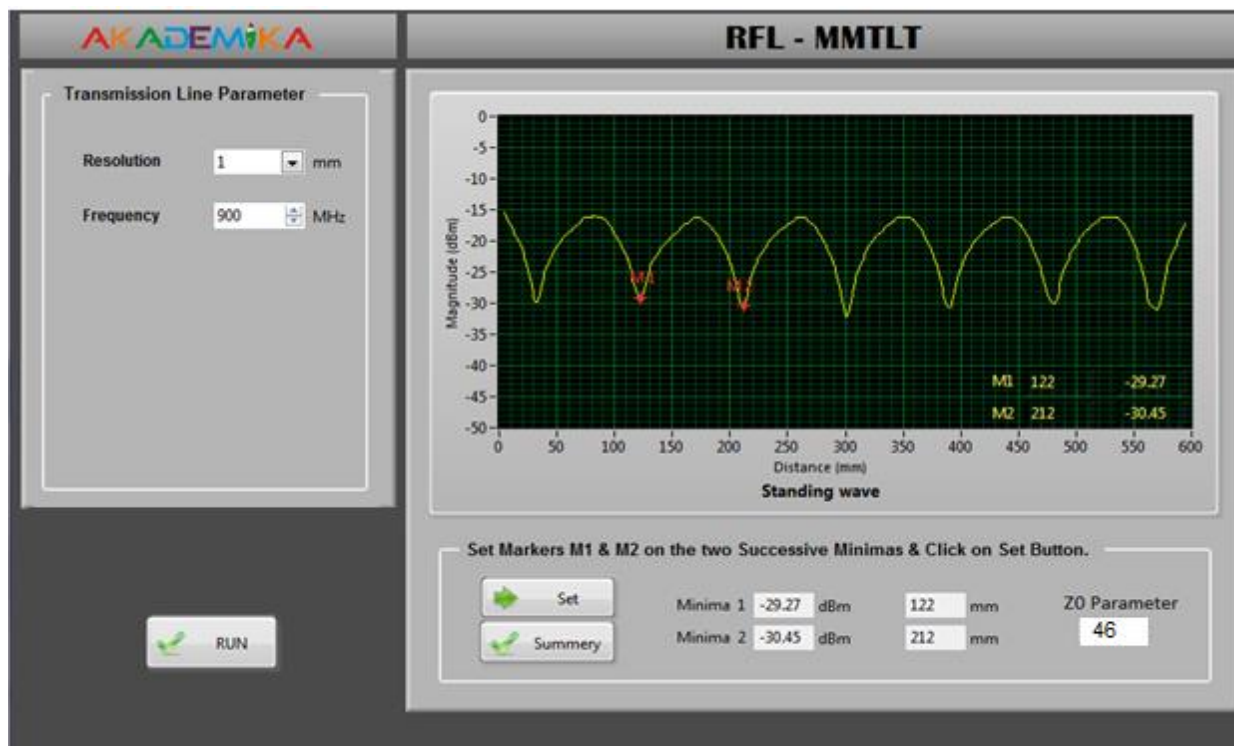
- **Calculation of $V_{\max}$**
 - $V_{\max} = \sqrt{P_{\max} * Z_0}$
- **Calculation of $V_{\min}$**
 - $V_{\min} = \sqrt{P_{\min} * Z_0}$
- **Calculation of VSWR**
 - $VSWR = \frac{V_{\max}}{V_{\min}}$

OPEN Condition

Follow the same procedure and replace the Port 3 of the Transmission Line JIG with OPEN termination condition and get the results

For OPEN Category Insert Z_0 parameter as 46 (Refer Annexure 1)

(Note: This is impedance ($Z_0=46$ of transmission line with OPEN termination which is measured on VNA at 900MHz frequency)



Press **Summary** to get all the transmission Line Parameters for your selected Load Condition OPEN termination



OBSERVATION TABLE FOR OPEN Condition

Frequency 900 MHz	1 st minima dBm	1 st maxima dBm	2 nd minima dBm	2 nd Maxima dbm	3 rd minima dBm	3 rd maxima dBm
Location of movable port (cm)						

STEP1

Consider any two adjacent minima and maxima

- $P_{\max} = 2^{\text{nd}} \text{ maxima (dbm)}$
- $P_{\min} = 2^{\text{nd}} \text{ minima (dbm)}$

(Consider any adjacent combination of minima and maxima.eg; - 2nd minima & 2nd maxima or 3rd minima & 3rd maxima)

STEP2

Convert to dbm to mW conversion

$$P_{\text{out}} (\text{mW}) = 1000 \times 10^{(\text{dBm} / 10)}$$

- $P_{\max} (\text{mW}) = 1000 \times 10^{(2^{\text{nd}} \text{ maxima} / 10)}$
- $P_{\min} (\text{mW}) = 1000 \times 10^{(2^{\text{nd}} \text{ minima} / 10)}$

STEP3

- **Calculation of $V_{\max}$**

- $V_{\max} = \sqrt{P_{\max} * Z_0}$

- **Calculation of $V_{\min}$**

- $V_{\min} = \sqrt{P_{\min} * Z_0}$

- **Calculation of VSWR**

- $VSWR = \frac{V_{\max}}{V_{\min}}$

OBSERVATION TABLE:-

LOAD/TERMINATION	OPEN	SHORT	LOAD
$V_{\min}$			
$V_{\max}$			
VSWR			
Load Impedance Z_L			

EXPERIMENT NO.3

EXPERIMENT NO: 3

NAME:-

Measurement of Incident and Reflected Wave in a Transmission Line

AIM:-

Concept of transmission and Plotting the Distance V/s Power Graph

EQUIPMENTS:-

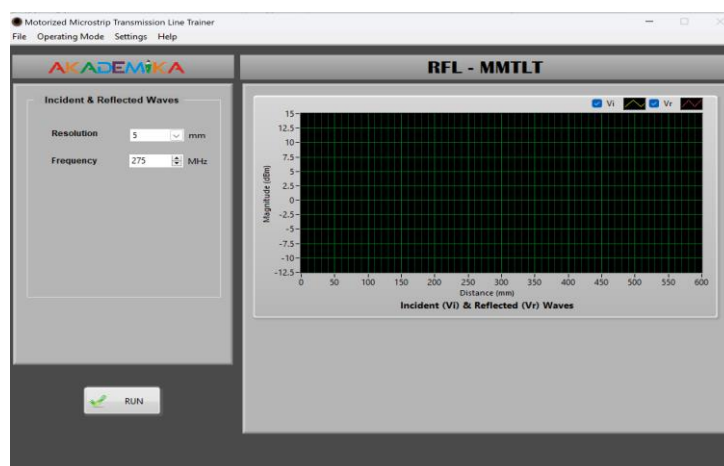
List of equipment's needed for performing the experiments

DUT, Equipment's and Accessories	Model Name	Quantity
Micro strip transmission line	Transmission Line JIG	1
RFL-MMTLT Control Unit	Control Unit	1
cables	SMA Cable -2meter	1
	SMA Cable – 1 meter	2
Termination	OPEN	1
Software	Transmission Line Software on PC	1

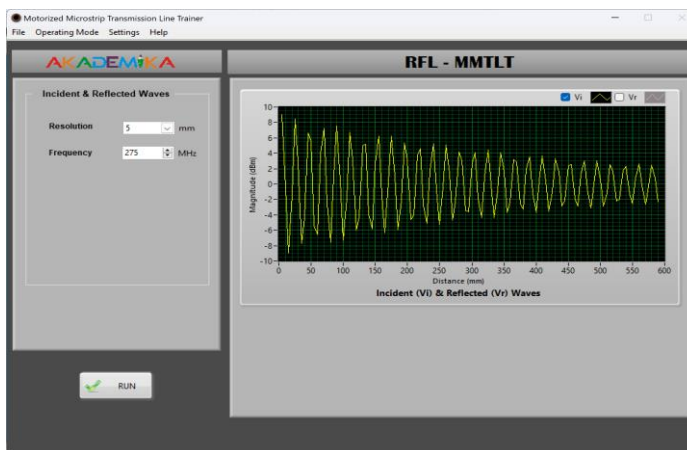
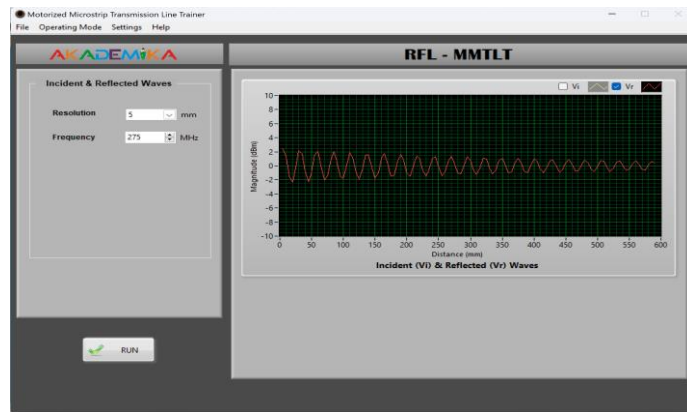
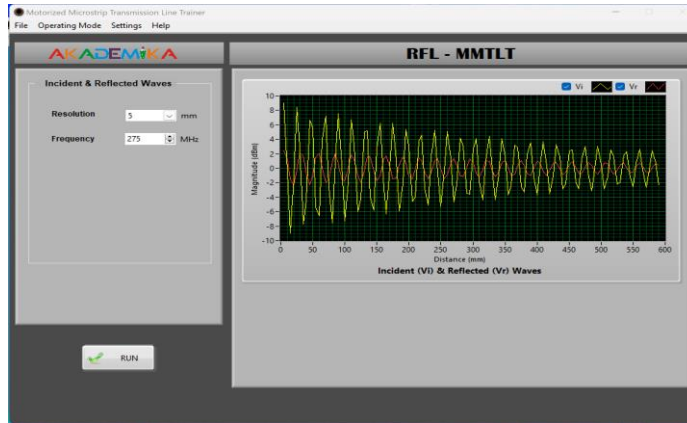
PC BASED CONTROL

PROCEDURE:

1. The Initial Connections for this experiment remains the same.
2. Terminate the Transmission Line with OPEN Termination at Port 3 of the Transmission Line JIG
3. Open the Software:
4. GO to Operating Mode - > Transmission Line - > Standing Wave - > Incident (V_i) & Reflected (V_r) Waves



5. Select Resolution : 5 mm and Frequency 275 MHz
6. Click the RUN Button on the Software
7. The Sensor probe will start moving along the Transmission line and after completion of the Travers it will display the Distance V/s Power graph.



8. The Yellow plot represents the Incident wave and Red plot represents the Reflected wave of the TL. You see that more Power approx (6 dBm) is reflected back to the source as the Line is OPEN.

Also you see that Power of the incident wave is reduced as the distance increases. In this case we see a power of (16dbm) at a distance of 50 mm and (5dbm) Power at 550mm

EXPERIMENT NO: 4

NAME:-

Experimentation for S-Parameters of Microstrip Transmission Line

OBJECTIVE:-

Determination of S-Parameters of Microstrip Transmission line.

AIM:-

Concept of transmission

- Calculation of 2 port S Parameters (S_{11} , S_{21} , S_{12} , S_{22})

EQUIPMENTS:-

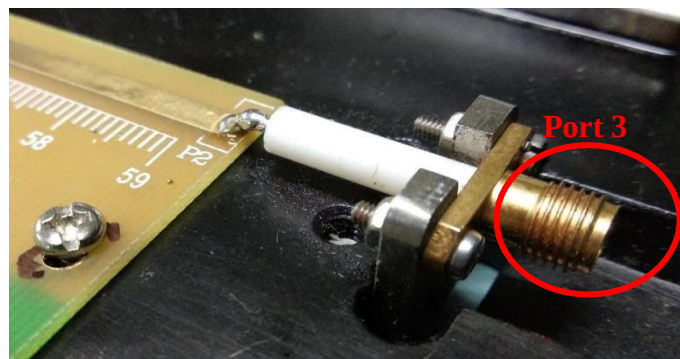
List of equipment's needed for performing the experiments

DUT, Equipment's and Accessories	Model Name	Quantity
Micro strip transmission line	Transmission Line JIG	1
RFL-MMTLT Control Unit	Control Unit	1
cables	SMA Cable -2meter SMA Cable – 1 meter	1 2
Directional Coupler	Directional Coupler	1
Software	Transmission Line Software on PC	1

PC BASED CONTROL

Hardware Connections for Insertion Loss (S_{21}) Measurement:

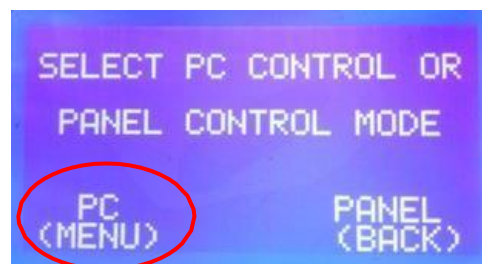
1. Connect the RF Source of Panel (**RF OUT**) to the **port 1** of the transmission line module using SMA 50 ohm 50cm Cable (i.e. signal generated by RFL-MMTLT RF Signal Generator is applied as input to the transmission line)
2. Connect the RF detector of Panel (**RF IN**) to the RFL-MMTLT **port 3** of transmission line assembly using SMA 50 ohm 2meter Cable



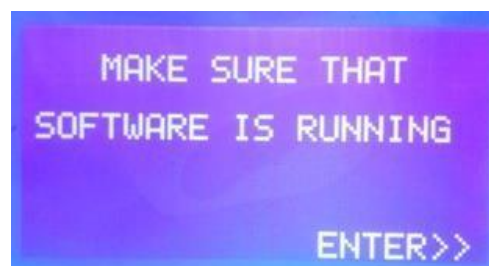
4. Do not connect anything at Port 2 (moving Probe).
 5. Do not connect anything at motor terminal Located on transmission line Jig.
 5. Turn on the RFL-MMTLT System, Apply Power using Power Cable and switch ON the Trainer Kit using Power ON switch present backside the trainer kit.
- Display will show following messages



6. Press **MENU** button on Front Control unit.
7. Select Control Mode. Press MENU key for PC control mode.
8. Connect the USB Cable between the Control Unit and PC.



9. Make sure that Software is running and press **ENTER**.



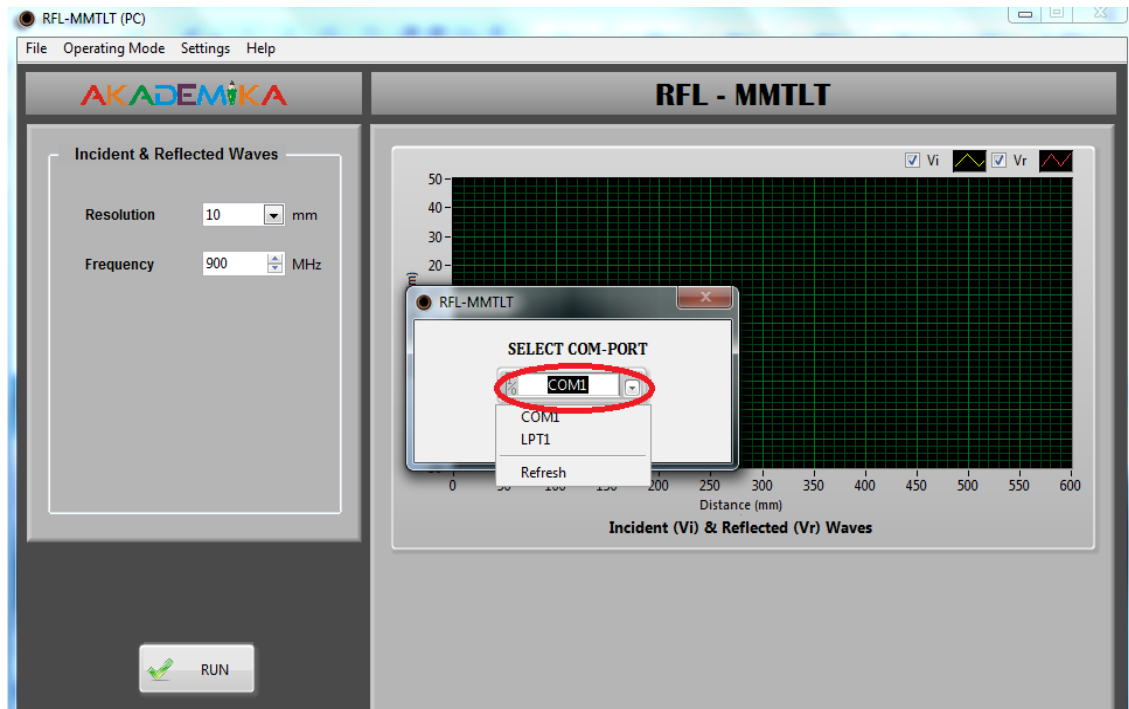
10. Display will show Software Control Mode Selected throughout the experiment..



Double click the RFL- MMTLT icon to Run the Software in PC



Select **COM PORT**: - Below picture shows the selection of com port. Click refresh button to update the COM port.



Procedure:-

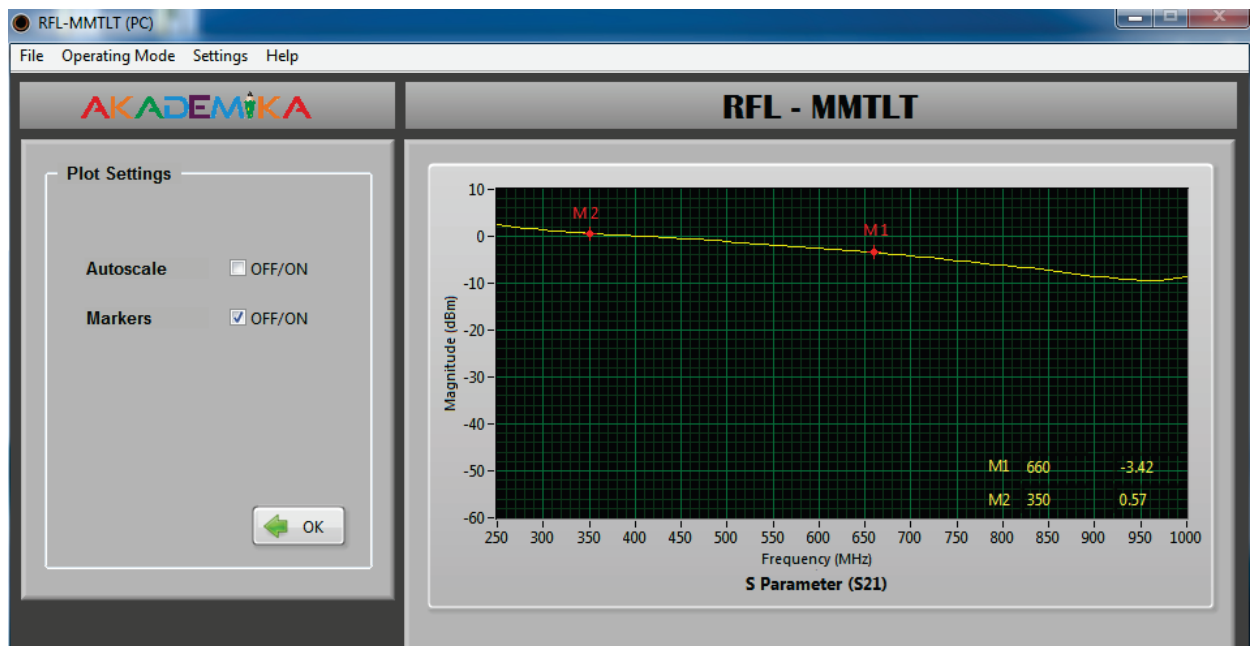
Set the RFL-MMTLT to below settings

1. Turn on the system
2. Select the PC mode

PC → OPEN GUI → SELECTS COM PORT → MENU BAR → OPERATING MODE → TRANSMISSION LINE → S-PARAMETERS (S21)

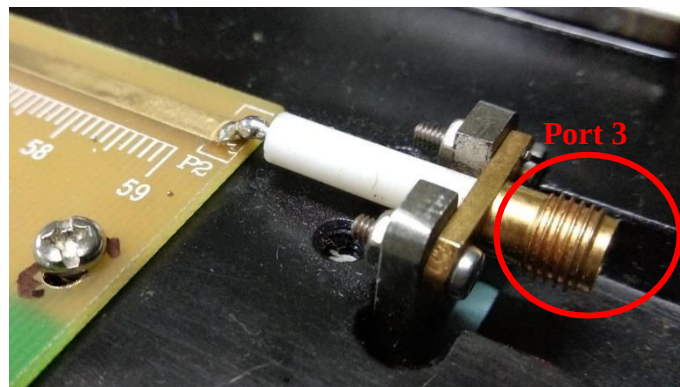
S21 Measurement

- Enter the start **250MHz** and stop **1000MHz** frequency for the Microstrip Transmission line.
- Run the system
- After completion of sweep of frequency, the graph appears on screen insert Markers and note down reading.



Hardware Connections for Insertion loss (S12) Measurement:

1. Connect the RF Source of Panel (**RF OUT**) to the **port 3** of the transmission line module using SMA 50 ohm 1 meter Cable (i.e. signal generated by RFL-MMTLT RF Signal Generator is applied as input to the transmission line)
2. Connect the RF detector of Panel (**RF IN**) to the RFL-MMTLT **port 1** of transmission line assembly using SMA 50 ohm 2 meter Cable



4. Do not connect anything at Port 2 (moving Probe).
5. Do not connect anything at motor terminal Located on transmission line Jig.

Procedure:-

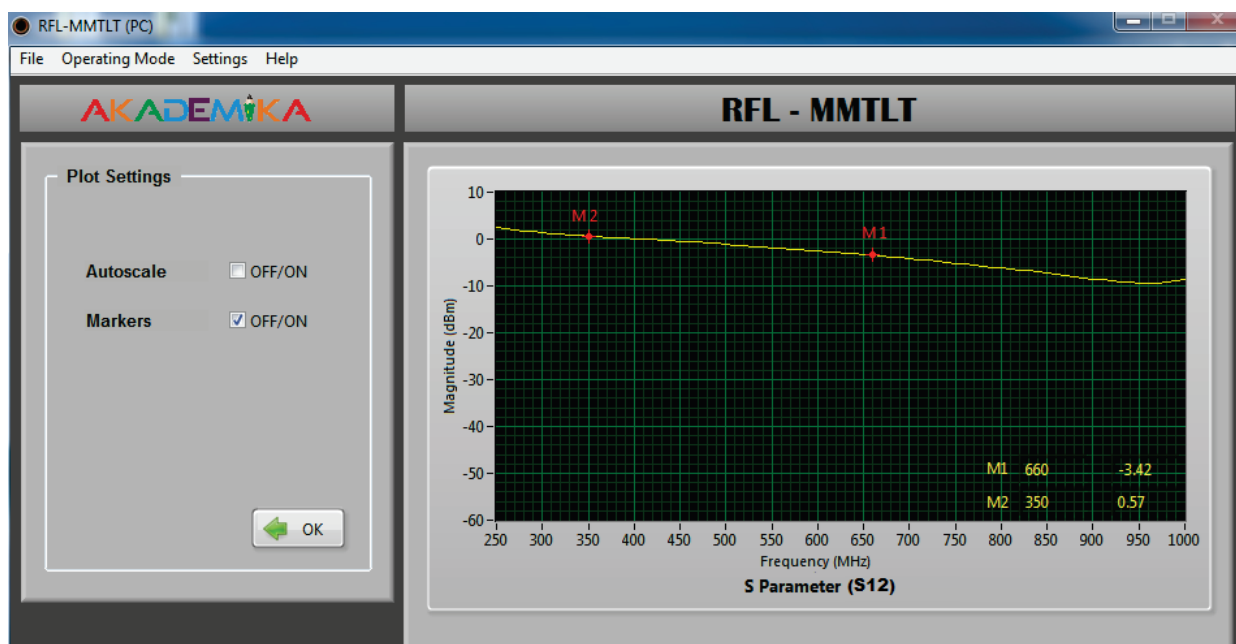
Set the RFL-MMTLT to below settings

- Turn on the system
- Select the PC mode

PC → OPEN GUI → SELECTS COM PORT → MENU BAR → OPERATING MODE → TRANSMISSION LINE → S-PARAMETERS (S12)

S12 Measurement

- Enter the start **250MHz** and stop **1000MHz** frequency for the Microstrip Transmission line.
- Run the system
- After completion of sweep of frequency, the graph appears on screen Insert Markers and note down reading.

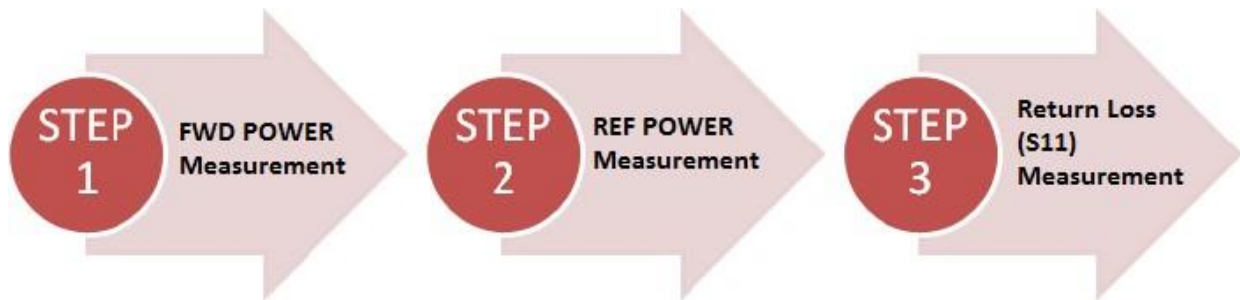


Hardware Connections for Scattering Parameters S11 and S22 Measurement:

SET-UP Required:

Item	RFL-MMTLT	SMA TO SMA Coaxial cable	PC with GUI	Three-port Directional coupler	Termination 50-Ohm
Ref Pic.					
Qty	1	3		1	1

FLOW Chart:-



Reflection (S11) Measurement:

Step 1. Forward power Measurement

1. “**RF OUT PORT**” of MMTLT panel is connected to **IN** of directional coupler using SMA 50cm coaxial cable.
2. “**RF IN PORT**” of MMTLT panel is connected to **CPL** of directional coupler using SMA 2meter coaxial cable.
3. The transmission line **Port1** is connect to **OUT** of directional coupler using SMA 50cm coaxial cable
4. Connect **50 Ohm** termination to transmission line **Port3**.

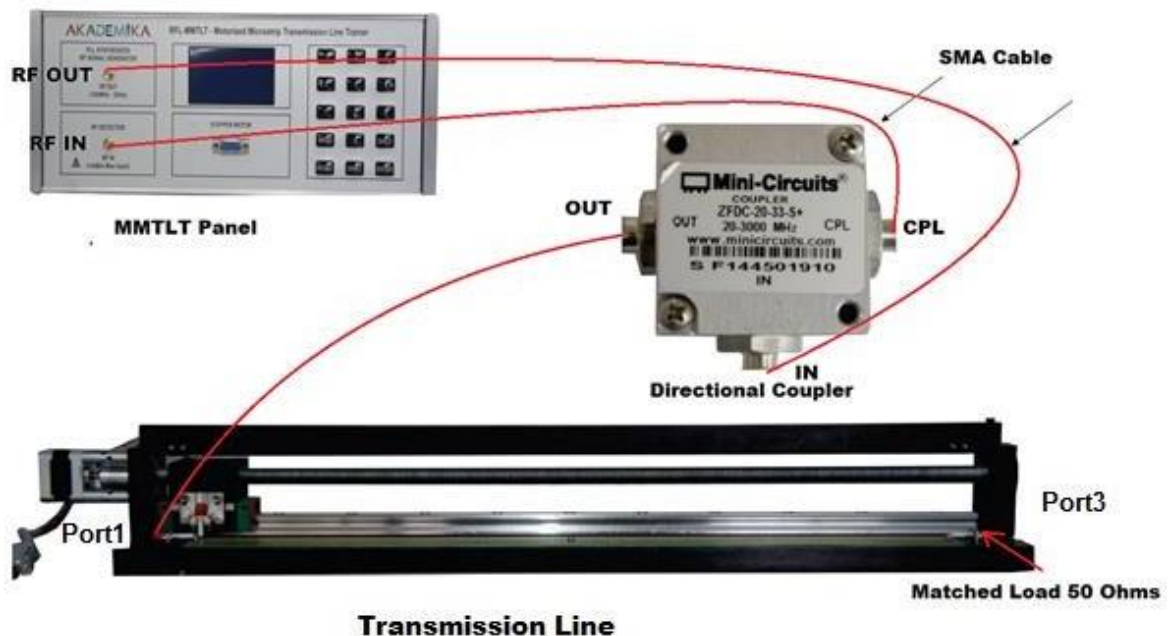


Fig.Measurement Connection Setup for Forward Power Measurement (S11)

PROCEDURE:-

Set the RFL-MMTLT to below settings

- Turn on the system
- Select the PC mode

PC → OPEN GUI → SELECTS COM PORT → MENU BAR → OPERATING MODE → TRANSMISSION LINE → S-PARAMETERS (S11)

1. Enter the start **250MHz** & stop frequency **1500MHz** for DUT Microstrip Transmission line.
2. Run the system.
3. After completion of sweep of frequency, the forward power will be stored in processor.
4. The next step software will show pop-up screen for reflected power measurement.

Step 2. Reflected power Measurement

5. “**RF OUT PORT**” of MMTLT panel is connected to **OUT** of directional coupler using SMA 50cm coaxial cable
6. The transmission line **Port1** connect to **IN** of directional coupler using SMA 50cm coaxial cable.
7. “**RF IN PORT**” of MMTLT panel is already connected to **CPL OUT** of coupler using SMA cable. (don't change it)
8. Click **ok** to RUN the system (do not change frequency settings!!)
9. After completion of sweep of frequency, the reflected power will be stored in processor and then software will show pop up screen the return loss graph.

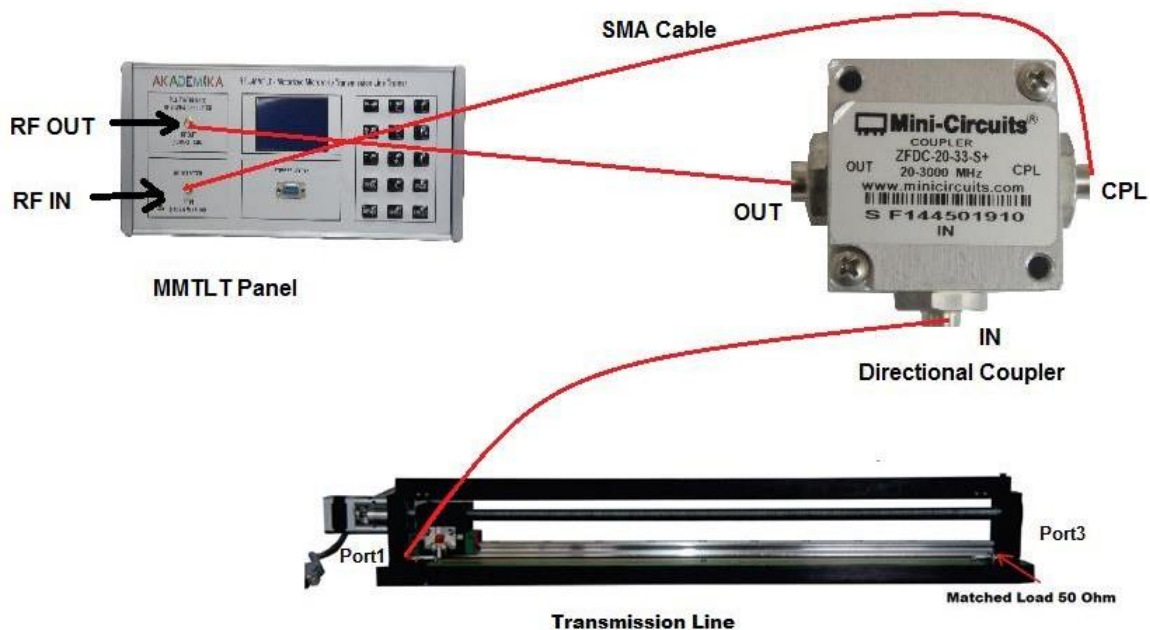


Fig. Measurement Connection Setup for Reflected Power Measurement

Step3. RETURN LOSS (S11) Measurement

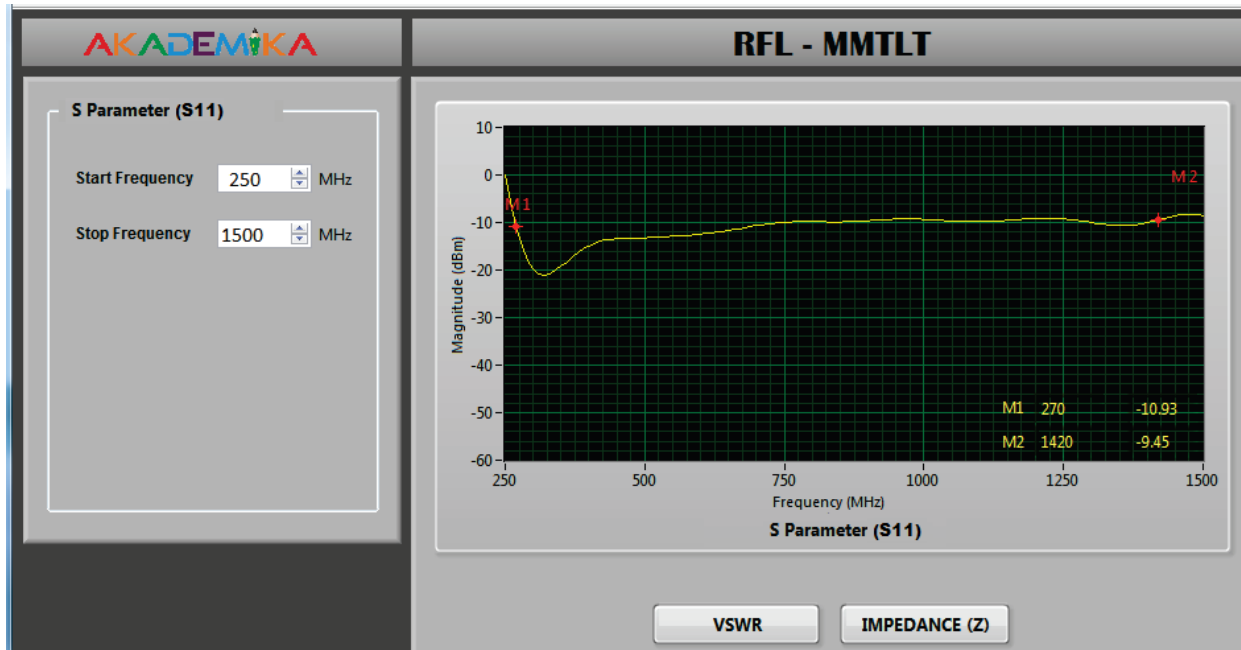
After successful completion of step 8, the software will show the return loss (S11) graph.

10. Return Loss graph calculated using Forward & Reflected power measurement ratio.

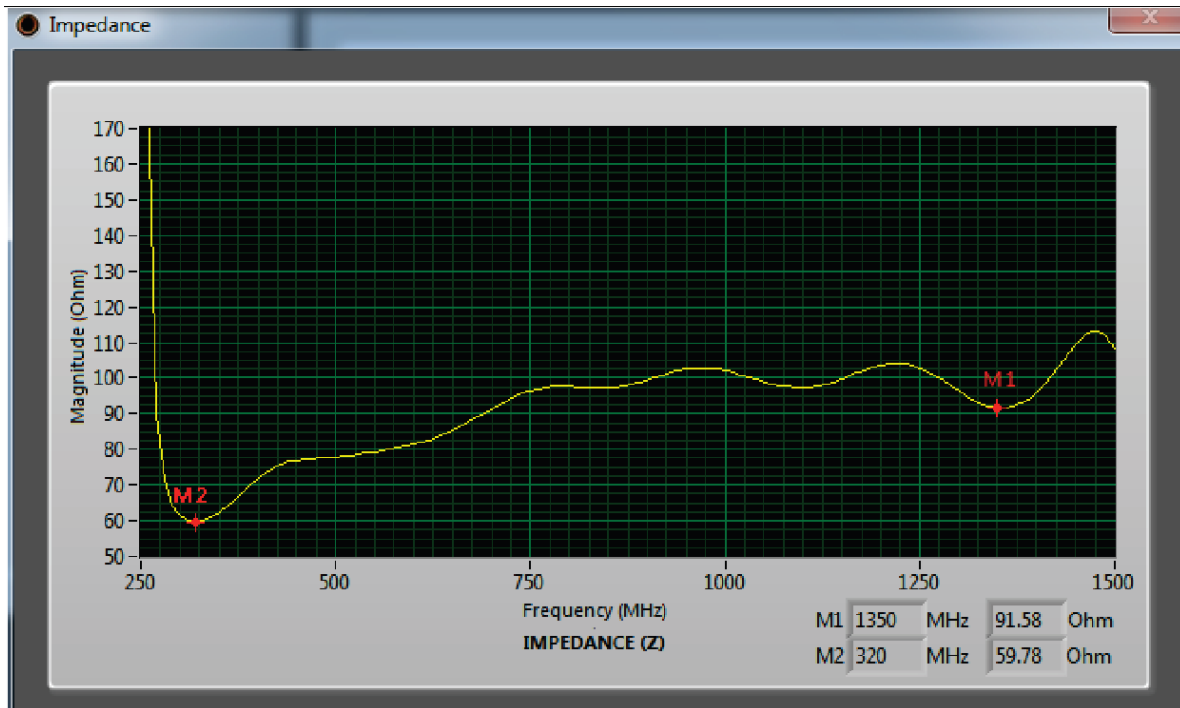
11. The graph appeared on the screen is frequency Vs power (s11).

12. To Plot VSWR and Impedance Graph Click on VSWR and Impedance option.

- After completion of sweep of frequency, the graph appears on screen Insert Markers and note down reading.



13. To Plot Impedance Graph, Click on Impedance option. Below impedance graph display



Reflection (S22) Measurement:

Step 1. Forward power Measurement

1. “**RF OUT PORT**” of MMTLT Panel is connected to **IN** of directional coupler using SMA 50cm coaxial cable
2. The transmission line **Port3** is connect to **OUT** of directional coupler using SMA 50cm coaxial cable.
3. “**RF IN PORT**” of MMTLT panel is connected to **CPL** of directional coupler using SMA 2m cable.
4. Connect **50 Ohm** termination to transmission line **Port1**.

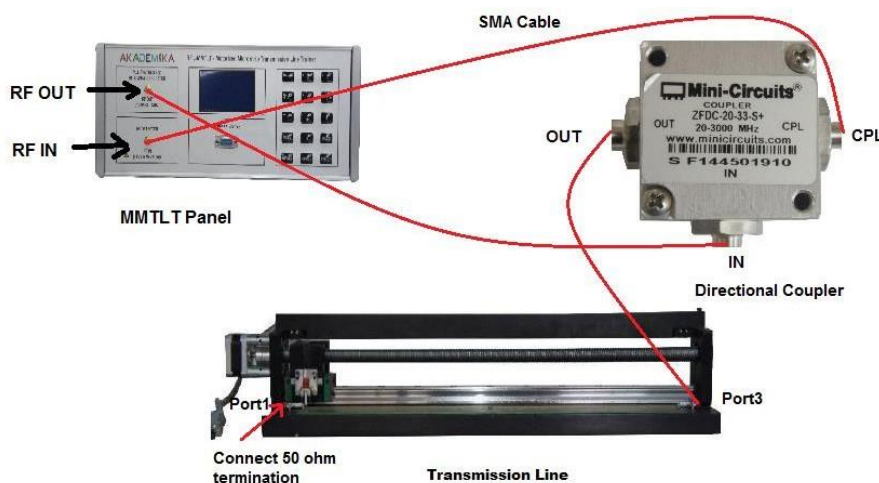


Fig.Measurement Connection Setup for Forward Power Measurement (S22)

PROCEDURE:-

Set the RFL-MMTLT to below settings

- Turn on the system
- Select the PC mode

PC → OPEN GUI → SELECTS COM PORT → MENU BAR → OPERATING MODE → TRANSMISSION LINE → S-PARAMETERS (S22)

1. Enter the start **250MHz** & stop frequency **1500MHz** for DUT Microstrip Transmission line.
2. Run the system.
3. After completion of sweep of frequency, the forward power will be stored in processor and then software will show pop-up screen for reflected power measurement (Step 2).

Step 2. Reflected power Measurement

4. “**RF OUT PORT**” of MMTLT panel is connected to **OUT** of directional coupler using SMA 50cm coaxial cable.
5. The transmission line **Port3** is connect to **IN** of directional coupler using SMA 50cm cable.
6. “**RF IN PORT**” of MMTLT panel is already connected to **CPL OUT** of coupler using SMA cable. (don't change it)
7. Click **ok** to RUN the system (do not change frequency settings!!)
8. After completion of sweep of frequency, the reflected power will be stored in processor and then software will show pop up return loss graph.

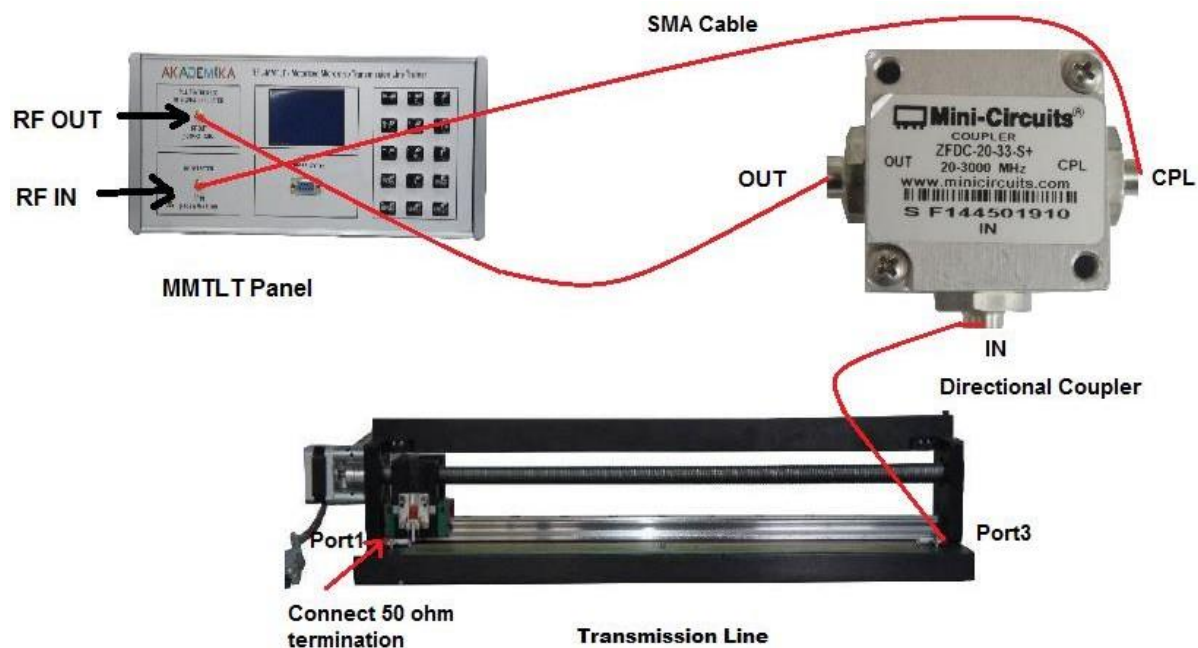
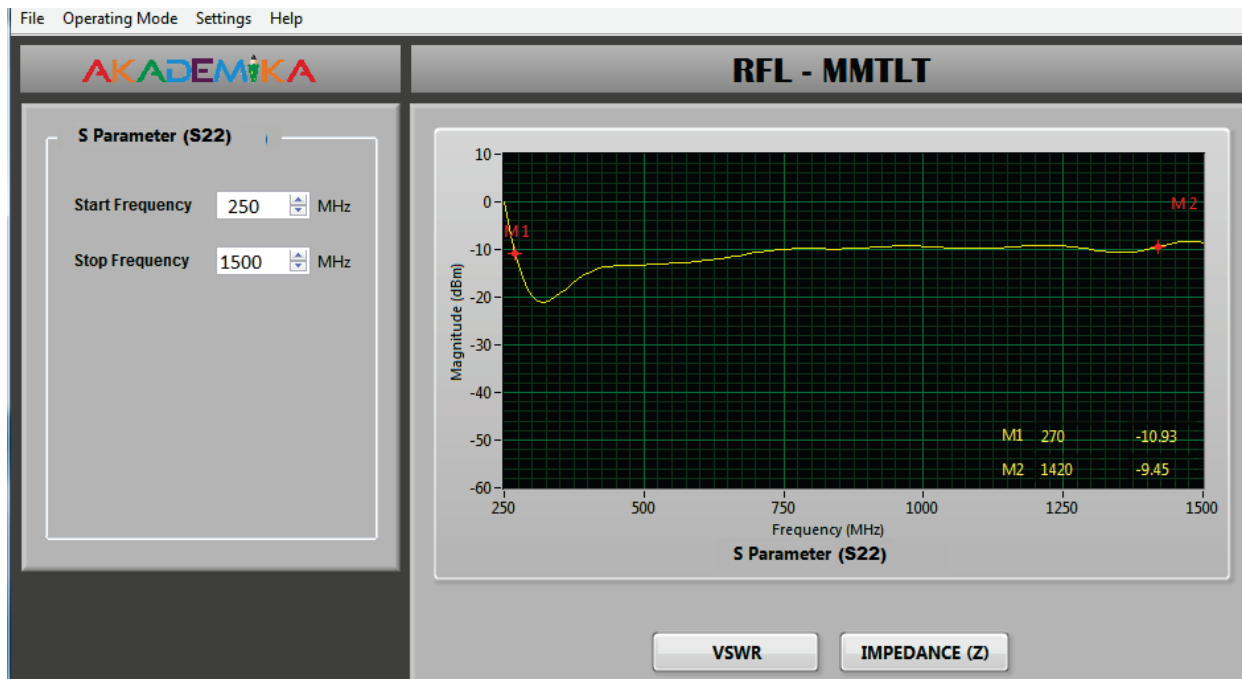


Fig.Measurement Connection Setup for Reflected Power Measurement (S22)

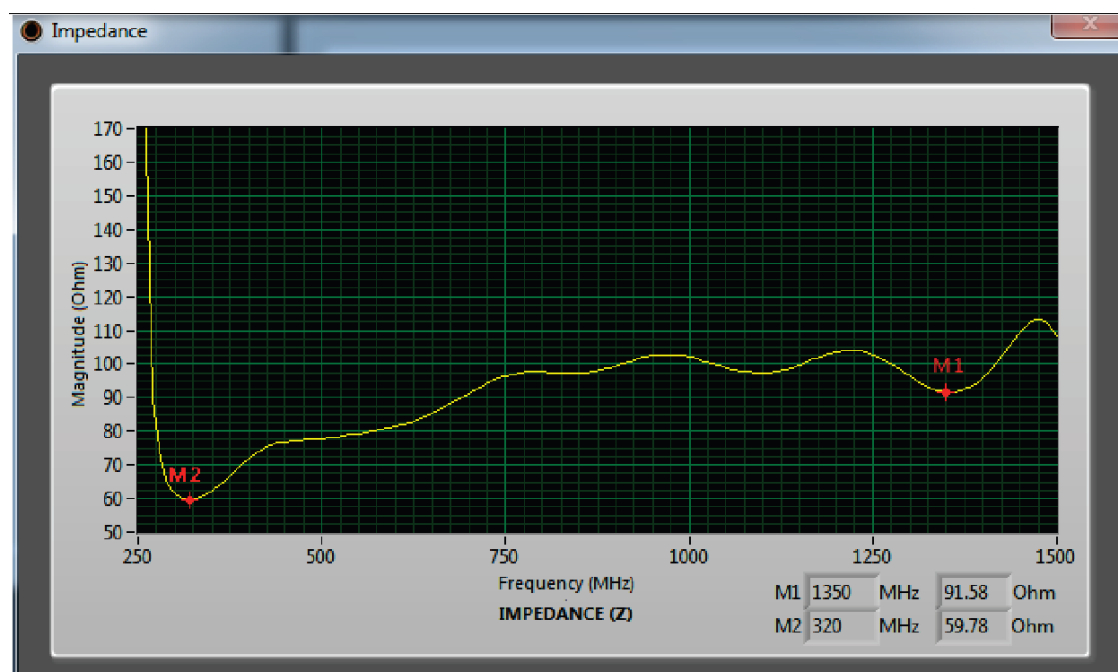
Step3. RETURN LOSS (S11) Measurement

After successful completion of step 8, the software will show the return loss (S11) graph.

9. Return Loss graph calculated using Forward & Reflected power measurement ratio.
10. The graph appeared on the screen is frequency Vs power (s11).
11. To Plot VSWR and Impedance Graph Click on VSWR and Impedance option.
 - After completion of sweep of frequency, the graph appears on screen Insert Markers and note down reading.



12. To Plot Impedance Graph, Click on Impedance option. Below impedance graph display



EXPERIMENT NO.5

EXPERIMENT NO: 5

NAME:-

Determination of unknown frequency using transmission line

OBJECTIVE:-

Determination of the unknown frequency

AIM:-

Determination of guided wavelength.

Determination of frequency using minima method.

EQUIPMENTS:-

List of equipment's needed for performing the experiments

DUT, Equipment's and Accessories	Model Name	Quantity
Micro strip transmission line	Transmission line JIG	1
RFL-MMTLT Control Unit	RFL-MMTL Control Unit	1
Cables	SMA Cable – 1 meter	1
	SMA Cable -2meter	1
	9 Pin D Type Male to Female Cable	1
	9 Pin D Type Male to DIN Connector Cable	1
Termination	Load	1
Software	Transmission Line Software on PC	1

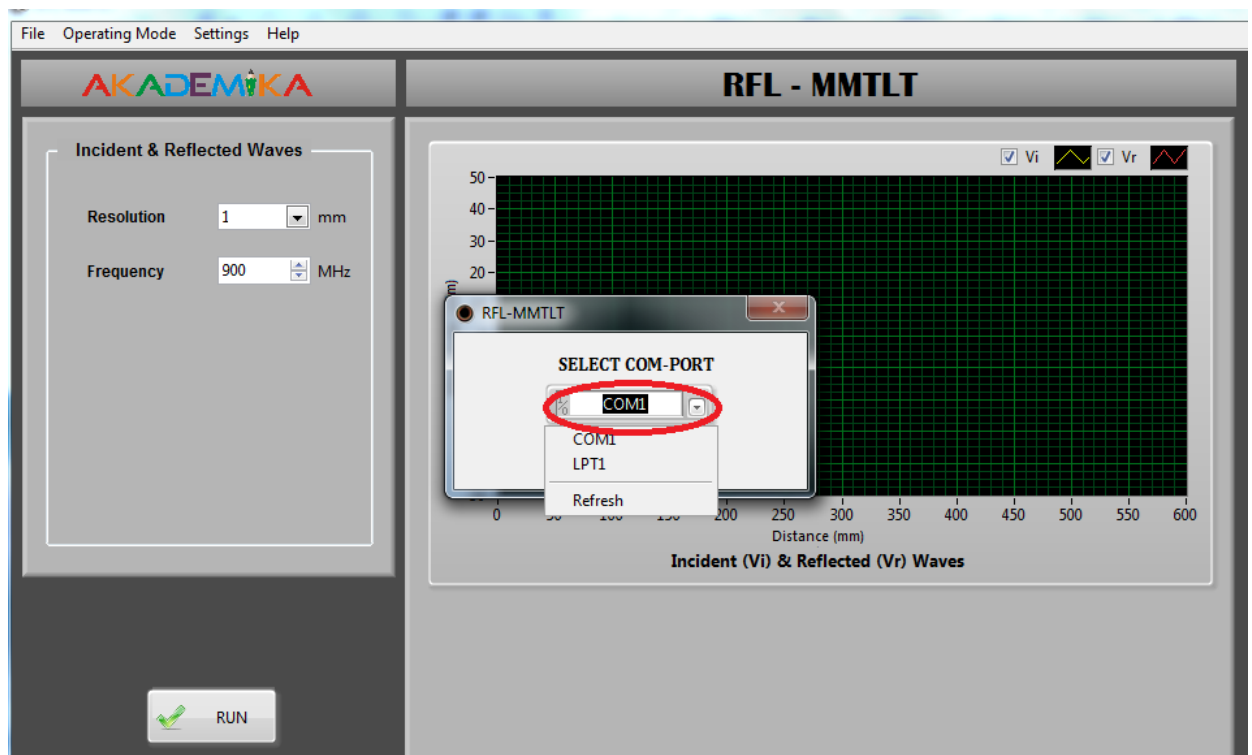
PC BASED CONTROL: -

Hardware Connections: -

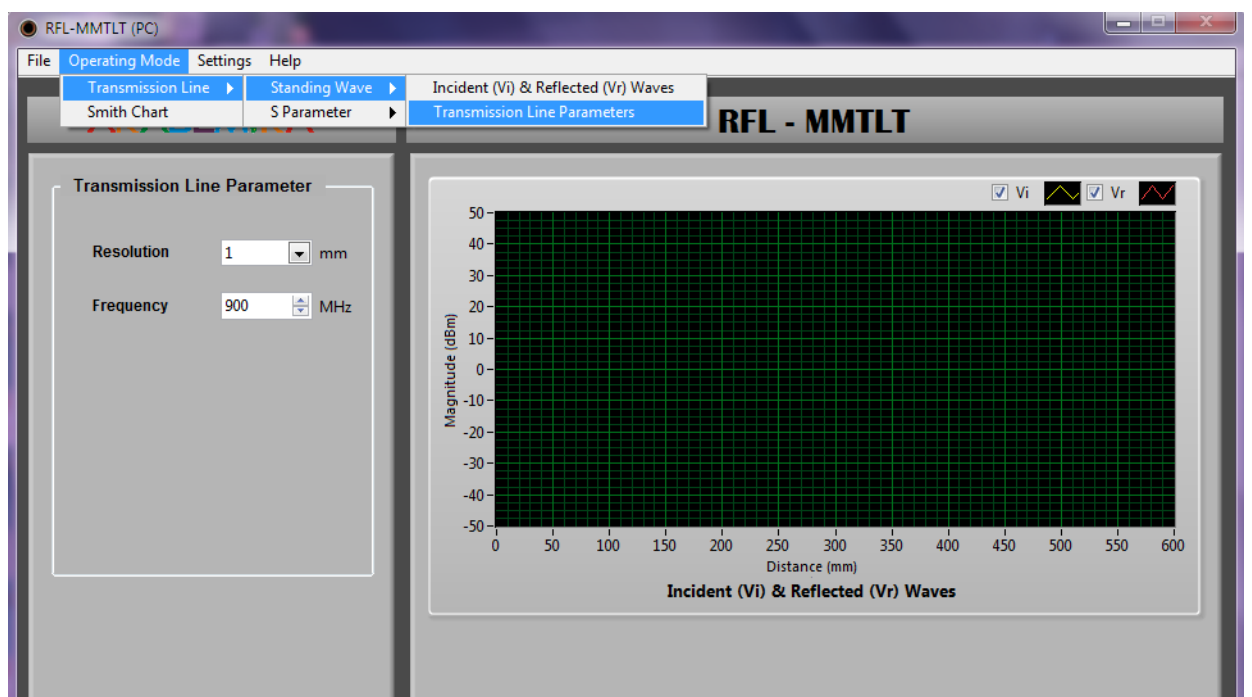
Make the Hardware Connections as per the Instructions given in Experiment No 1.

Double click the RFL- MMTLT icon to Run the Software in PC

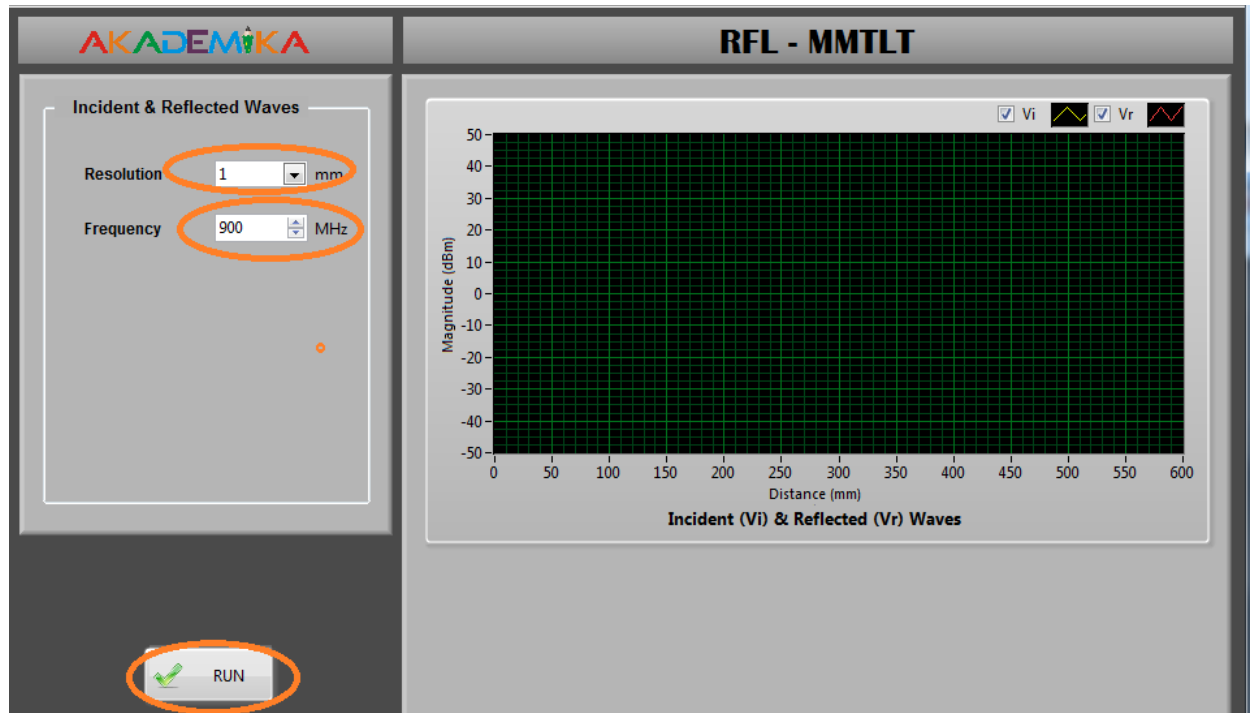




Select transmission line parameter as shown below



Select resolution and frequency 1mm & 900MHz respectively

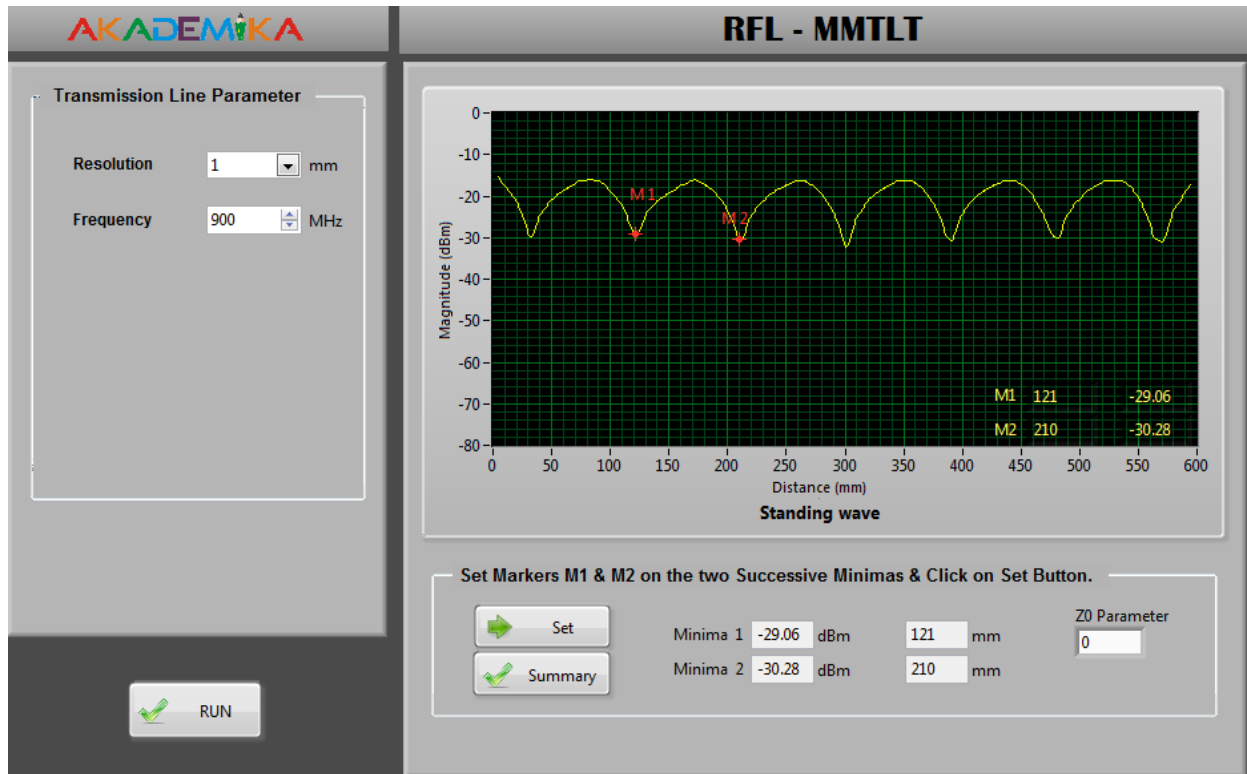


Click on RUN button to start experiment

- When we press RUN button motor starts running and signal sensing probe at port2 starts moving from 0cm of transmission line towards 59cm (Total 600mm). After completion of this distance moving probe returns back to 0cm position.

Note: During start of experiment, if moving probe position is not at 0cm, you can still RUN the experiment. It automatically comes to 0cm i.e. home position and then starts the experiment.

The sensor probe takes some minutes to complete its full rotation from generator to load. The graph will display on screen. Save the graph on computer.



- To save the image of wave go to **file>save as>Image File**
- To save the excel data of wave go to **file>save as>CSV File**
- Now repeat the same procedure to take readings for frequency 800MHz and 1000MHz
- Save multiple frequency waves in CSV for to particular location and the compare all the graphs
- To compare the graphs, go to **Settings>plot comparison**
- Browse the graphs you want to compare and observe the waves on software.

OBSERVATION TABLE

Frequency 1000 MHz	1 st minima dBm	1 st maxima dBm	2 nd minima dBm	2 nd maxima dBm	3 rd minima dBm	3 rd maxima dBm	4 th minima dBm
Location of movable port (cm)							

STEP1

Consider any two adjacent minima and maxima

- Location of movable port = 2nd minima (cm)
- Location of movable port = 3rd minima (cm)

(Consider any adjacent combination of minima or maxima.eg; - 2nd minima & 3rd minima or 2nd maxima & 3rd maxima)

Step2

Calculation of guided wavelength

$$\lambda_g = 2 * (2^{\text{nd}} \text{ minima} - 3^{\text{rd}} \text{ minima}) \text{ ----- cm}$$

STEP3

Calculation of unknown frequency

$$\text{Frequency} = \frac{c}{\lambda_g \sqrt{3.8}}$$

Step 4

Set the following specification & repeat the procedure

Observation Table: -

Given frequency (MHz)	900 MHz	800 MHz	1000 MHz
Calculated Frequency from transmission Line (MHz)			

EXPERIMENT NO.6

EXPERIMENT NO: 6

NAME:-

Plotting a Normalized Impedance on the Smith Chart

OBJECTIVE:-

Plotting a Normalized Impedance on the Smith Chart

AIM:-

Determination of Normalized Impedance.

Plotting Normalized Impedance of LOAD termination on Smith chart.

Plotting Normalized Impedance of OPEN termination on Smith chart.

EQUIPMENTS:-

List of equipment's needed for performing the experiments

DUT, Equipment's and Accessories	Model Name	Quantity
Micro strip transmission line	Transmission Line JIG	1
RFL-MMTLT Control Unit	RFL-MMTL Control Unit	1
cables	SMA Cable – 1 meter	1
	SMA Cable -2meter	1
	9 Pin D Type Male to Female Cable	1
	9 Pin D Type Male to DIN Connector Cable	1
Termination Load	LOAD	1
	OPEN	1
Software	Transmission Line Software on PC	1

PC BASED CONTROL

Hardware Connections:-

1. Set up the Hardware Connection as per Experiment No 1
2. Connect LOAD at Port 3 of transmission line assembly.
3. Turn on the RFL-MMTLT System, Apply Power using Power Cable and switch ON the Trainer Kit using Power ON switch present backside the trainer kit.

Procedure:-

Set the RFL-MMTLT to below settings

- Connect USB cable between MMTLT System and PC.
- Select the PC mode

PC → OPEN GUI → SELECTS COM PORT → MENU BAR → OPERATING MODE → TRANSMISSION LINE → TRANSMISSION LINE PARAMETERS

Load Impedance (ZL) Measurement

- Enter the frequency 900MHz and Resolution 1mm
- Run the system
- After completion of sweep of frequency, the graph appears on screen set two minima Markers on graph and Click on **Summary**

Note down reading of ZL at 900MHz.

- Similarly measured ZL at 800MHz and 1000MHz by changing frequency and run software
- Run system again.

Observation Table: -

Sr. No	Frequency (MHz)	Load Impedance (ZL in ohms)	Normalized Impedance (Znorm) (ohms)
1.	800MHz		
2.	900MHz		
3.	1000MHz		

Calculation of Normalized Impedance

$$Z_L / Z_0 = Z_{\text{norm}}$$

Plotting a Normalized Impedance on the SMITH CHART

1. First, normalize the impedance as follows:

For Example, at 900MHz freq $Z_L=54$ ohm and $Z_0=50$ ohm

$$Z_L / Z_0 = Z_{\text{norm}}$$

$$Z_{\text{norm}} = 54/50 = 1.08 \text{ at } 900\text{MHz}$$

Similarly, you can calculate at 800MHz and 1000MHz

Putting this Z norm values in first row in software which is called Real Part (R)

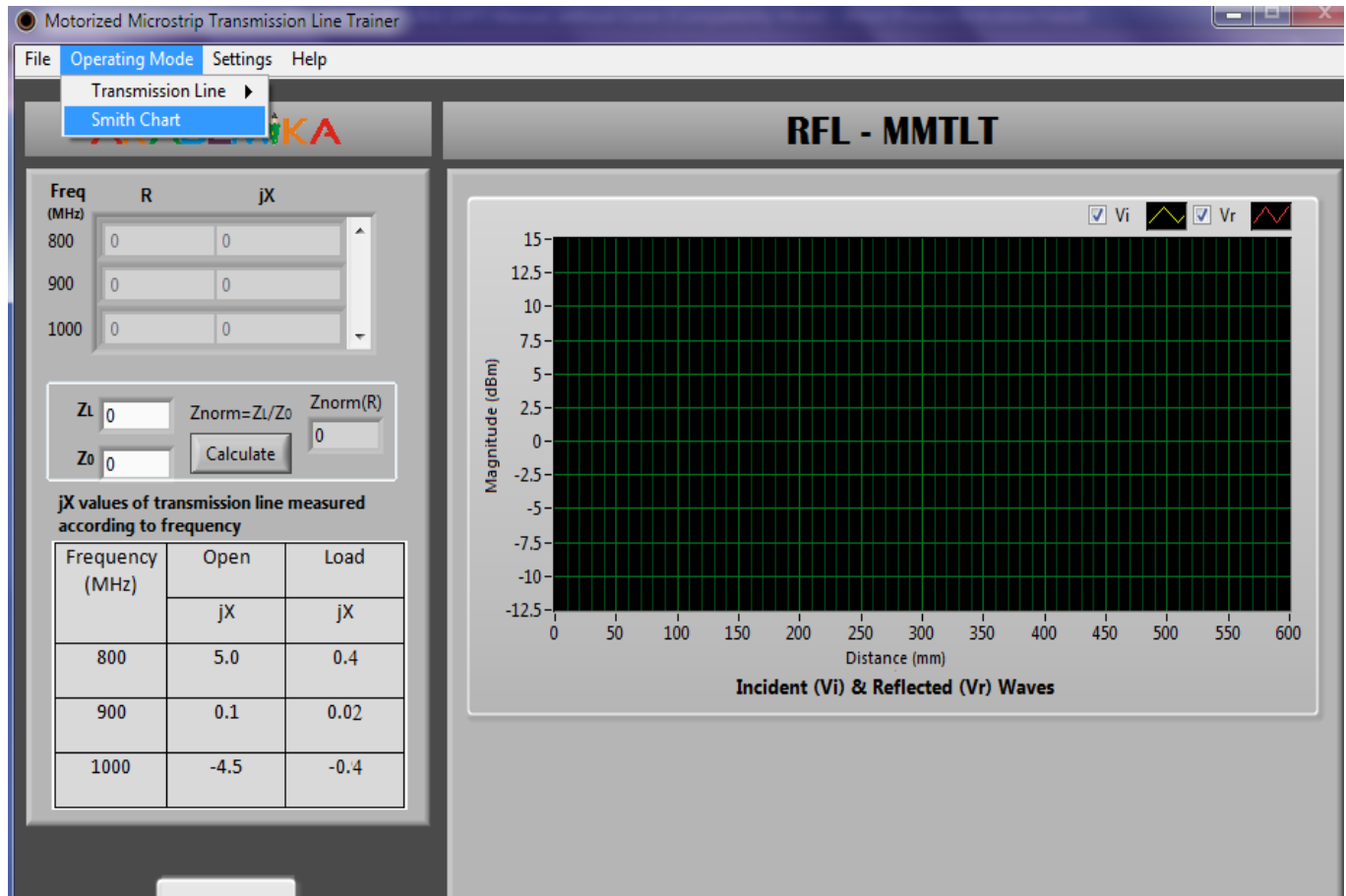
Then putting R values at 800MHz, 900MHz, 1000MHz in 1, 2 and 3 row of R column receptively.

2. Imaginary Part (jx) of Transmission Line as measured on VNA at 800MHz, 900MHz and 1000MHz.

Then putting jx values at 800MHz, 900MHz, 1000MHz in 1, 2 and 3 row of jx column receptively.

Which shown in table for LOAD condition.

SELECT MENU → OPERATING MODE → SMITHCHART



Now putting Normalize Impedance(R) values in first row and Imaginary values(jx) is second row as follow and click on run to plot smith chart graph

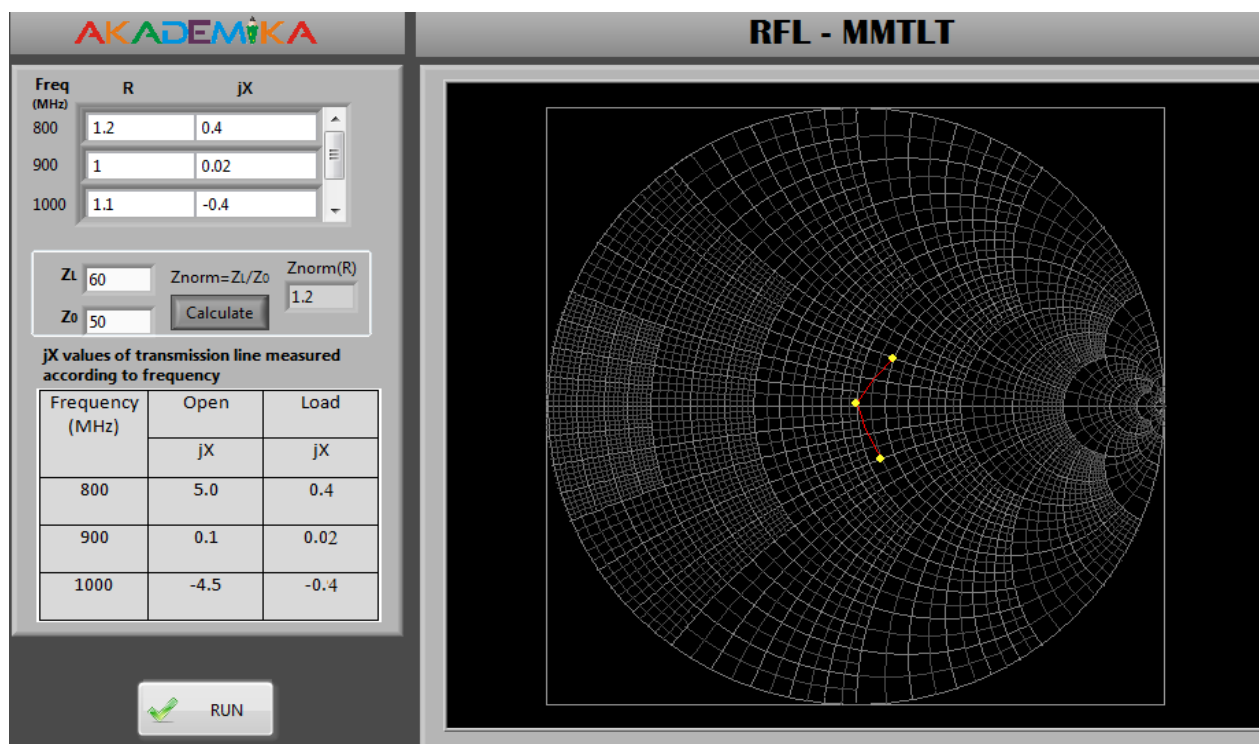


Fig.1. Plotting Normalized Impedance for load termination

OPEN Category Normalized Impedance Plotting

Follow the same procedure as per Open category except step number 4. Here at Port 3 Connect OPEN termination condition with different frequency and get the results for ZL

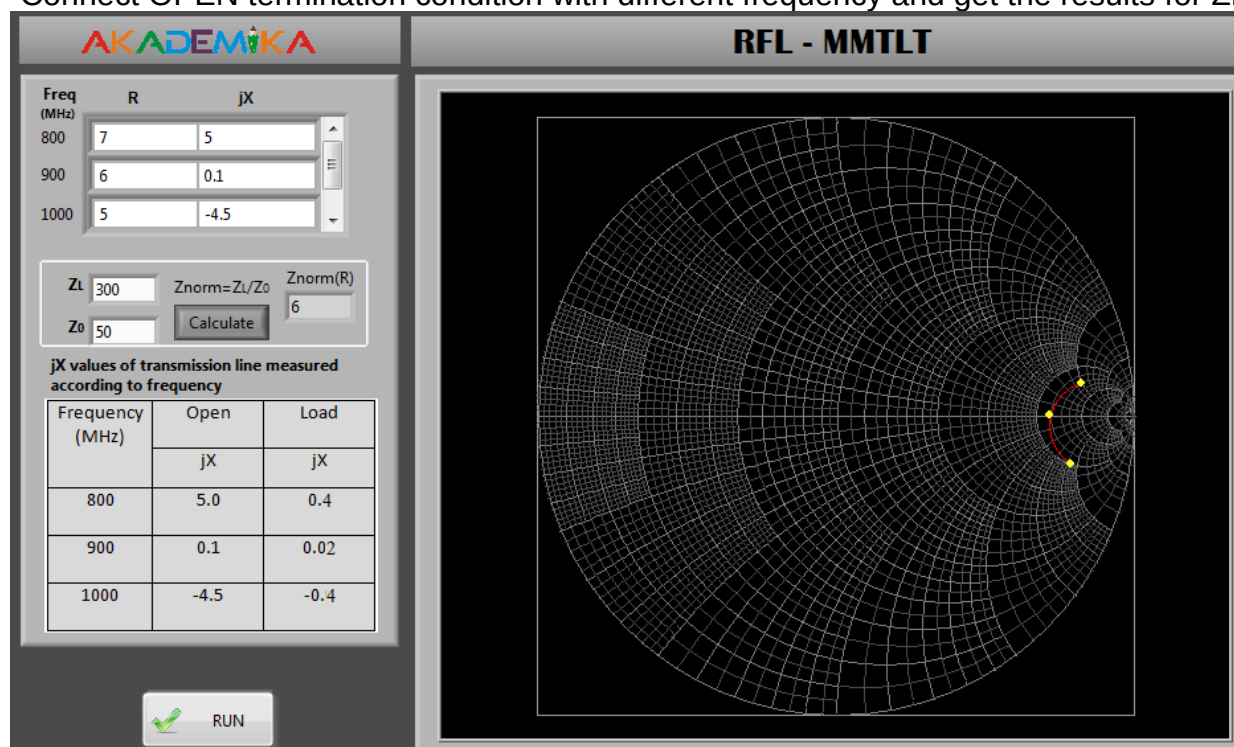


Fig.2. Plotting Normalized Impedance for open termination

Annexure 1

1) For LOAD Termination Z_0

Sr. No	Freq (MHz)	Impedance(Z_0) (ohms)
1.	800	35
2.	900	28
3.	1000	29

2) For SHORT Termination Z_0

Sr. No	Freq (MHz)	Impedance(Z_0) (ohms)
1.	800	4
2.	900	3
3.	1000	9

3) For OPEN Termination Z_0

Sr. No	Freq (MHz)	Impedance(Z_0) (ohms)
1.	800	86
2.	900	46
3.	1000	38

Note: This is impedance values (Z_0) of transmission line with LOAD, SHORT and OPEN termination which is measured on VNA at different frequencies.